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Master's thesis

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**ACADEMIC RESEARCH IN QUANTUM COMPUTING,
QUANTUM KEY DISTRIBUTION AND POST-QUANTUM
CRYPTOGRAPHY**

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ABSTRACT

English

The study analyzed quantum, post-quantum, and quantum key cryptography (QKD) algorithms, with an emphasis on the implementation and performance of the Shor's and Grover's algorithms. They were compared with their classical counterparts, assessing their potential applications in the face of modern technological limitations.

The research showed the theoretical superiority of quantum algorithms, especially Grover's in searching unstructured data and Shor's in number factorization. However, current technological limitations mean that in practice, modern cryptographic systems are not at risk at this very moment.

The paper highlights both the potential and current limitations of quantum computers, offering valuable lessons for future research and development of quantum technologies. The results obtained can provide a solid basis for further research into improving algorithms and accelerating technological progress in this field.

Key words: Quantum Mechanics, Quantum Computer, Quantum algorithm, Post-Quantum algorithm, Quantum Key Distribution (QKD), Shor's algorithm, Grover's algorithm.

Polish

W pracy przeanalizowano algorytmy kwantowe, post-kwantowe oraz kwantową kryptografię klucza (QKD), z naciskiem na implementację i wydajność algorytmów Shora i Grovera. Porównano je z klasycznymi odpowiednikami, oceniając ich potencjalne zastosowania w obliczu współczesnych ograniczeń technologicznych.

Badania wykazały teoretyczną przewagę algorytmów kwantowych, zwłaszcza Grovera w przeszukiwaniu nieustrukturyzowanych danych oraz Shora w faktoryzacji liczb. Jednak obecne ograniczenia technologiczne sprawiają, że w praktyce współczesne systemy kryptograficzne nie są na razie zagrożone.

W pracy podkreślono zarówno potencjał, jak i obecne ograniczenia komputerów kwantowych, oferując cenne wnioski dla przyszłych badań i rozwoju technologii kwantowych. Uzyskane wyniki mogą stanowić solidną podstawę do dalszych badań nad udoskonaleniem algorytmów i przyspieszeniem postępu technologicznego w tej dziedzinie.

Słowa kluczowe: Mechanika kwantowa, komputer kwantowy, algorytm kwantowy, algorytm post-kwantowy, kwantowa dystrybucja klucza, Algorytm Shora, Algorytm Grovera.

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1. Introduction

The future is happening here and now. Are the secrets stored on the Internet still safe? Are the logins, passwords, accounts, or even digital signatures used and will they remain protected? Can they be cracked, stolen, or even bypassed in a matter of minutes, contrary to existing knowledge and methods? Will the cryptographic systems, encryption protocols, authentication mechanisms used so far become obsolete and insufficiently secure in the face of rapidly developing quantum algorithmics? Is there already a demand for alternative methods?

These and many other questions are answered by this scientific study. It touches on the biggest opportunities and threats that humanity will face today and in the coming years in the field of Cybersecurity.

The dynamic development of quantum computers represents a breakthrough in computing, promising unprecedented computing power and capabilities that go beyond the limitations of classical computers. This development has and will continue to have an impact in a wide variety of fields, including cryptography, where the emergence of quantum algorithms threatens the security of current cryptographic systems. To gain an in-depth understanding of the opportunities and threats posed by the above changes, it is necessary to approach the subject in two ways. Even playing the role of both white-hat and black-hat hackers. With the goal of securing information in the quantum age, a key area of this academic research has become an attempt to break through the previously known and used classical methods, implementing and testing quantum computing (QC) algorithms on a publicly available quantum computer, not forgetting the topic of quantum attack resistant cryptography (PQC), as well as over quantum key distribution (QKD).

Primarily, this paper focuses on the current state, mode of operation and progressive development of quantum computers, in the context of the quantum algorithms used on them, the possibilities they offer, and the risks associated with them. It touches on their advantages, advantages over the classical cryptography known and used so far, both delving into the sphere of theoretical foundations and their practical implementation, comparing the possibilities they offer with previously known methods, not forgetting the current challenges in these areas.

Secondly, it addresses the topic of quantum key distribution and cryptography resistant to quantum attacks. By analyzing recent developments, studying countless scientific articles, and identifying potential areas for further research, this work aims to contribute to the development of robust cryptographic solutions that can confront the evolving threats from quantum computers. The ultimate goal is to make sure that the digital world remains secure and resilient in the face of this transformative technological revolution.

Equally importantly, the research conducted, the resulting outcomes and the thoughts presented are intended as a complex compendium of knowledge, both for those who are able to draw conclusions from the results achieved and make further use of them, and for those who are just beginning their adventure with quantum concepts, or even simple classical cryptography. This work was intended to clearly introduce interested people to the subject, explaining all issues from the ground up and gradually developing them through the research conducted, comparisons and their presented quantifiable results.

This topic is little known and only developing. It requires intensive promotion, advocacy, and education. Currently, it is treated marginally and often downplayed, despite its potential to revolutionize modern technology and significantly impact the security of the digital world. Because it is a relatively new topic, college curricula typically do not cover it or address it as a curiosity and a vision of the future, forcing students and researchers to educate themselves in the field. In addition, problems in finding the latest research and scientific publications in this field, due to their currently small number, hinder the deepening of knowledge and the development of innovative solutions. Therefore, it is necessary to increase the awareness and bring attention of the academic and professional community to the topics of quantum and post-quantum computing and quantum computers to realize their full potential.

1.1. Purpose and layout of the thesis

The work consists of two main parts: theoretical and research. The theoretical part discusses the history of the development of quantum mechanics, quantum algorithms, quantum computers and post-quantum algorithms. It also includes an explanation of the theoretical foundations, explaining in an accessible way the key issues of quantum mechanics, as well as the principles of operation of quantum computers and algorithms. The content has been formulated in a way that is understandable to a wide audience, which makes the work accessible not only to specialists, but also to people less familiar with the topic.

The research part focuses on the practical implementation of Shor's and Grover's algorithms using the Qiskit library package on a quantum computer powered by a one hundred twenty-seven qubits Eagle processor from IBM. An uncountable number of experiments were conducted to assess the efficiency of these algorithms in comparison with their classical counterparts. During the research, numerous problems and limitations related to the current state of quantum technology were encountered, but despite this, both algorithms were successfully implemented and evaluated.

Creating a master's thesis in Cybersecurity, focusing on the topic of cryptography, algorithms and quantum computers has multiple advantages and is particularly valuable for several key reasons:

- **Revolutionary technology**

Quantum computers and quantum algorithms represent a breakthrough in computing. Working in this field allows researchers to be at the forefront of a new wave of technology that has the potential to transform many aspects of information technology.

- **Threats to current cryptographic systems**

Quantum algorithms, such as Shor's algorithm, may in the future threaten current cryptographic systems based on the difficulty of factoring large numbers (for example RSA). Research in this area, in particular the implementation of quantum algorithms and their testing, confirms these reports, the evidence of which is described in the following chapters. Further considerations on this topic are necessary and may contribute to the development of new, quantum-resistant cryptographic systems.

- **New methods of security**

In addition to the threats, quantum computers offer new security methods, such as quantum key distribution (QKD), which provides unconditionally secure communication. An equally interesting topic and worth studying.

- **Interdisciplinarity and skill development**

This subject combines elements of computer science, physics, mathematics, and cryptography, which allows the development of a wide range of skills and knowledge. The mentioned difficulty, resulting from the poor base of scientific articles in the present field, provides wide opportunities to act and join the ranks of pioneers in promoting knowledge in this field. There is a growing demand for specialists with knowledge of quantum computers and their applications in cyber security.

- **Impact on the future of cyber security**

The goal of this research is to contribute to the development of new strategies and technologies for securing data in the face of threats from quantum computers. The opportunity to contribute to information protection in the future is extremely motivating and rewarding.

1.2. Division of tasks

The division of tasks in the creation of this master's thesis was crucial to ensuring its versatility and comprehensive development of the topic. A joint decision was made among the authors to assign Michał Olejnik responsibility for the theoretical, substantive, and editorial side. This allowed for a detailed discussion and explanation of both complex concepts and phenomena related to quantum mechanics and computers, quantum algorithms and their implementations, and post-quantum algorithms, as well as issues related to the history of this field. He also took care of the graphical representation of the obtained results and the experiments discussed in this thesis. Mateusz Świercz focused on the practical side, in particular on the implementation and testing of algorithms in the IBM Quantum environment, using the Qiskit library package. In addition, he was responsible for obtaining and maintaining contact with research laboratories and experts in the field of quantum technology, which enabled obtaining the necessary technical information and practical assistance in the implementation of advanced research projects. However, this would not have been possible without the mutual support that was the foundation of this cooperation. Regular consultations and joint problem-solving allowed for the smooth overcoming of encountered difficulties. The creators of this work contributed their unique skills and knowledge, which enabled a comprehensive approach to research and its practical application. Thanks to this collaboration, everyone could focus on their strengths, while complementing and supporting each other where necessary, which significantly improved the quality of the result of this work.

Such division of tasks was necessary to combine theoretical knowledge with practical experience and obtain complete research results.

1.3. Acknowledgments

This work encountered several significant challenges. Many of them resulted from both current technological limitations and the lack of new articles and scientific sources in this field, from which one could develop one's knowledge and which would provide reliable substantive support. The variability and lack of compatibility of individual versions of the Qiskit library package additionally hindered the achievement of the intended goals.

In this context, cooperation with representatives of the Poznan Supercomputing and Networking Center (PSNC) proved particularly important. Thanks to their expert knowledge and commitment, it was possible to solve technical and substantive problems, which significantly contributed to the successful completion of the work.

2. Quantum computing - Historical Background

It might seem that the subject of quantum mechanics and the factors and phenomena resulting from it, extending classical mechanics, is relatively new, but the beginnings of this issue date back to 1900.

The following chapter focuses on presenting the most notable events, the introduction of which was necessary to conduct a reliable and in-depth analysis of the topic. Its existence is necessary to understand the context and significance of the fields discussed in this work. It introduces concepts and terms that are developed and explained throughout the work. However, its purpose is only to bring closer the most notable events because a thorough description of each of them, together with a detailed background, could become a separate, extensive work.

This chapter has been divided into chronologically arranged events, which were then assigned to appropriate categories. The presentation of the historical development of these fields allows for understanding their mutual connections and significance for modern science and technology.

2.1. Quantum mechanics

Quantum mechanics is a branch of physics that deals with the description of the behavior of matter and energy at the level of atoms and elementary particles. It is the foundation of quantum theory and technology, and knowing its history allows readers to better understand how the concepts and theories that today form the basis of quantum computers developed [1].

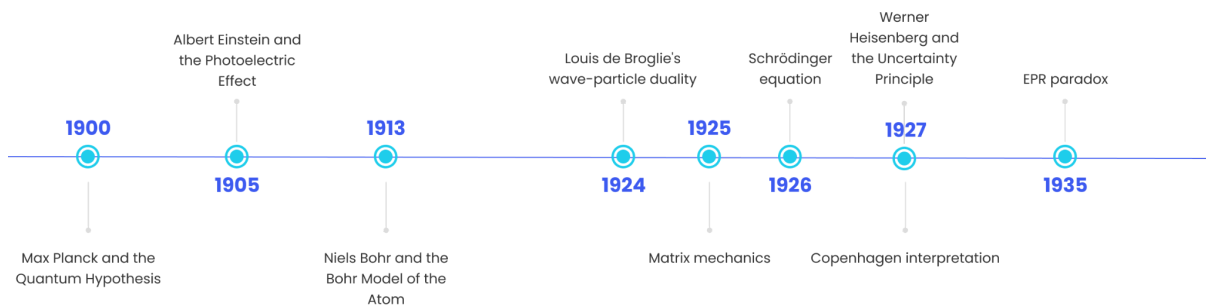


Figure 2.1: History of Quantum Mechanics (Created by the Author)

2.1.1. 1900: Max Planck introduced the concept of energy quanta, which initiated the development of quantum mechanics [2].

Historical context

In the 19th century, despite the rapid development of technology and industry, many physicists believed that the field of physics was almost completely understood, leaving only a few minor problems to be solved. One of these problems was the radiation of heated bodies, which could not be explained by the known laws of classical physics.

“Ultraviolet catastrophe”

Attempts to describe the radiation of heated bodies using classical physics, based on the laws of thermodynamics and wave theory, such as the Rayleigh-Jeans law, led to unreachable results, suggesting that the energy emitted by a ideal black body increases infinitely with increasing frequency of radiation, especially in the ultraviolet region. This led to the absurd conclusion that the total energy emission by such a body should be infinite, which was contrary to observations. This problem was called the "ultraviolet catastrophe".

Planck’s breakthrough

Max Planck, after five years of intensive research, proposed a revolutionary solution to this problem. He found that energy is not emitted continuously, but in discrete portions, which he called quanta. In December 1900, Planck presented his equation describing blackbody radiation, which was consistent with experimental results and eliminated the problem of infinite energy.

Black body - “a hypothetical body that absorbs without reflection all of the electromagnetic radiation incident on its surface and then, depending on temperature, emits its own thermal radiation” [\[64\]](#).

Due to the informational nature of this chapter, equations, formulas, and their explanation will not be introduced here.

The significance of this discovery

Planck's discovery started a new era in physics – quantum mechanics. His law became the foundation for all modern electronics, including computers, smartphones, and satellites. Quantum mechanics influenced the development of many fields of science and technology, changing the perception of the world at the subatomic level. It started the development of research in this field, which can be observed to this day.

2.1.2. 1905: Albert Einstein explained the photoelectric effect by introducing the concept of light quanta (photons) [\[3\]](#).

In 1905, Albert Einstein published a paper on the photoelectric effect, explaining that light consists of particles called photons, each of which carries a certain, discrete amount of energy proportional to its frequency. Einstein's work on the photoelectric effect provided significant support for Planck's quantum hypothesis and contributed to a better understanding of the particle nature of light.

2.1.3. 1913: Niels Bohr and the Bohr Model of the Atom [\[4\]](#):

Bohr's model of the atom assumed that electrons orbit the nucleus in well-defined, quantized energy levels. This model provided a deeper understanding of atomic structure, as it proposed that the energy of electrons is not continuous but rather exists in discrete amounts. Bohr's theory provided crucial insights into the nature of energy within atoms, helping to explain phenomena that could not be accounted for by earlier models.

Additionally, Bohr introduced the concept of quantum jumps, a groundbreaking idea that described how electrons could transition between these quantized energy levels by either absorbing or emitting photons. This concept not only reinforced the idea of energy quantization but also played a vital role in explaining the spectral lines observed in atomic emissions and absorptions. The introduction of quantum jumps highlighted the profound significance of quantization in atomic systems, further solidifying Bohr's contributions to the development of quantum mechanics.

2.1.4. 1924: Louis de Broglie proposed wave-particle duality, suggesting that matter particles have wave properties [\[5\]](#).

This theory states that subatomic particles, such as electrons and photons, exhibit both particle and wave properties. This means that, depending on the experiment, they can behave like material particles with a defined mass and position, or like waves with interference and diffraction properties. This duality has been confirmed in numerous experiments, such as the double slit experiment.

Double-slit experiment (Young's interference experiment)

In the classical double slit experiment, photons exhibit interference when they are not observed, demonstrating their wave nature. However, when detectors are placed at the slits to record which slit a photon passes through, the interference disappears, and the photons begin to behave like particles. This illustrates how the act of measurement affects the state of the photon, according to the uncertainty principle. This experiment was also cited and discussed in the following chapters.

2.1.5. 1925: Matrix mechanics (Werner Heisenberg, Max Born and Pascual Jordan) [\[6\]](#).

It was the first complete and consistent description of quantum mechanics. It is an extension of the Bohr Model, describing how quantum jumps occur. In this theory, physical quantities such as position and momentum, as well as physical quantum systems are represented by matrices, not by traditional continuous variables. On the other hand, quantum states, which describe the full information about the quantum system, for example through properties such as entanglement, superposition, uncertainty, are described by vectors.

2.1.6. 1926: Schrödinger equation [\[7\]](#).

The Erwin Schrödinger equation is a key element of quantum mechanics, describing the evolution of a particle's quantum states. There are two main forms of this equation:

- Time-independent: Describes the stationary states of a particle in each potential, specifying possible energy values and the corresponding wave functions.
- Time-dependent: Describes how the particle's wave function changes over time, allowing predictions of the dynamics of a quantum system.

Both forms of the Schrödinger equation are fundamental for predicting the behavior of particles at the quantum level, both in terms of their energy and the time evolution of quantum states.

2.1.7. 1927: Uncertainty principle (Heisenberg's indeterminacy principle) [\[8\]](#).

In 1927, Werner Heisenberg, a German physicist, introduced the uncertainty principle, a fundamental concept in quantum mechanics. The principle states that it is impossible to simultaneously determine both the exact position and momentum of a particle; the more precisely one is known, the less precisely the other can be measured. This principle also emphasizes how the act of measurement disturbs the quantum state of a particle, such as a photon. Heisenberg's uncertainty principle underscored the intrinsic limitations of observing quantum systems and played a crucial role in the later development of quantum computing.

2.1.8. 1927: Copenhagen interpretation (Niels Bohr, Werner Heisenberg) [\[9\]](#).

The Copenhagen interpretation, developed mainly by Niels Bohr and Werner Heisenberg, is one of the most important approaches to understanding quantum mechanics. It assumes that quantum reality cannot be described in a classical way and that our language and intuition are insufficient to fully grasp its essence. Instead, it focuses on a pragmatic approach, recognizing that the key element is the act of measurement, which causes the collapse of the particle's wave function into a specific state.

Key elements of the Copenhagen interpretation:

- Complementarity: Wave and particle descriptions are complementary, and both are necessary for a complete description of quantum phenomena.
- Wave Function Collapse: The wave function describing the state of a particle collapses to a specific outcome during the act of measurement.
- Intuitive constraint: Quantum mechanics requires abandoning classical intuition and accepting abstract mathematical concepts.

“We must be clear that when it comes to atoms, language can be used only as in poetry. The poet, too, is not nearly so concerned with describing facts as with creating images and establishing mental connections.” ~ Niels Bohr [\[63\]](#).

“The concepts of classical physics form the language by which we describe the arrangements of our experiments and state the results. We cannot and should not replace these concepts by any others. Still the application of these concepts is limited by the relations of uncertainty. We must keep in mind this limited range of applicability of the classical concepts while using them, but we cannot and should not try to improve them.” ~ Werner Heisenberg [\[65\]](#).

“We wish to speak in some way about the structure of atoms and not only about the ‘facts’ (...) But we cannot speak about the atoms in ordinary language.” ~ Werner Heisenberg [\[66\]](#).

The above quotes are the credo of the Copenhagen approach. There are things difficult to understand using everyday language based on intuition and common classical mechanics, and perhaps never will be understood.

2.1.9. 1935: Introduction of EPR paradox [\[10\]](#).

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen challenged the conventional understanding of quantum mechanics by publishing a paper that introduced what would later be called the Einstein–Podolsky–Rosen (EPR) paradox. This paradox brought to the forefront concerns about the completeness of quantum mechanics, highlighting the enigmatic phenomenon of **quantum entanglement**, which is described in detail in the next chapter.

The EPR paradox suggested that if quantum mechanics were complete, it would lead to seemingly paradoxical situations in which two or more particles, once entangled, would have their properties so deeply intertwined that measuring the state of one particle would instantly determine the state of the other, even over vast distances. This instantaneous, faster-than-light “spooky action at a distance,” as Einstein famously described it, seemed to violate the principle of locality, a cornerstone of classical physics. In this regard, Einstein, Podolsky, and Rosen argued that quantum mechanics may not be a complete theory, and that there may be hidden variables yet to be discovered that could explain these nonlocal correlations.

The publication of the EPR paradox started an intense debate in the scientific community, questioning the very foundations of quantum theory. It raised profound questions about the nature of reality and the limits of human knowledge in describing the physical world. Although the EPR paradox did not disprove quantum mechanics, it revealed counterintuitive implications of the theory and suggested that our understanding of the quantum world may be incomplete.

2.2. Quantum and Post-Quantum Algorithms

Quantum cryptography

Quantum algorithms, such as Shor and Grover's algorithm, have revolutionized the approach to computation, showing the potential of quantum computers to solve problems inaccessible to classical computers. These are topics that were later given their own chapters, because they are the main topic of this dissertation. For this reason, they appear here out of obligation to place them hierarchically in history, but all the details are described in chapters number 4 and from 7 to 11.

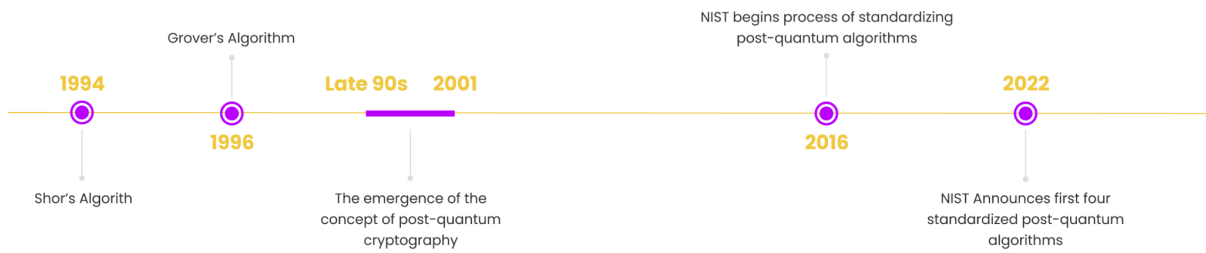


Figure 2.2: History of Quantum and Post-Quantum Algorithms (Created by the Author)

2.2.1. 1994: Peter Shor developed Shor's algorithm [\[11\]](#).

It is a quantum algorithm that solves the problem of factoring large integers into their prime factors in time exponentially faster than any known classical algorithm. Its development was a breakthrough because it showed that quantum computers could potentially break many current cryptographic systems, such as RSA, which rely on the difficulty of factoring large numbers. Shor's algorithm became one of the main motivators for intensive research and development of quantum computers.

2.2.2. 1996: Lov Grover presented Grover's algorithm [\[12\]](#).

It is a quantum algorithm that significantly speeds up searching unstructured databases. While classical algorithms require on average $O(N)$ operations to find a search item in a database of size N , Grover's algorithm does it in $O(\sqrt{N})$ operations, a quadratic improvement over classical method. This discovery, like Shor's algorithm, demonstrated the potential of quantum computers to solve problems much faster than classical computers, with major implications for the fields of optimization and cryptography.

Post-Quantum cryptography

2.2.3. Late 90s & 2001: The concept of post-quantum cryptography is created [\[21\]\[22\]](#).

In response to Shor's results, researchers began to think about the future of cryptography in the era of quantum computers. Although quantum computers did not yet exist in practice, the vision of their development forced scientists to look for new cryptographic solutions that would be resistant to quantum attacks.

In 2001, the concept of post-quantum cryptography began to take formal shape. Researchers realized that it was necessary to develop new algorithms that would be secure even in the face of advanced quantum computers. The term "post-quantum cryptography" began to be used in the scientific literature to describe these new techniques.

2.2.4. 2016: NIST (National Institute of Standards and Technology) has begun the process of standardizing post-quantum algorithms [\[23\]](#).

In 2016, the National Institute of Standards and Technology (NIST) announced a competition to develop and standardize new cryptographic algorithms that would be resistant to quantum attacks. This was a significant step in the development of post-quantum cryptography, which focused the world's attention on the need to create new security standards. The competition attracted numerous proposals from around the world, announcing the results in July 2022

2.2.5. 2022: NIST Announced First Four Cryptographic Algorithms That Will Be Post-Quantum Cryptography Standards [\[23\]](#).

“The announcement follows a six-year effort managed by NIST, which in 2016 called upon the world’s cryptographers to devise and then vet encryption methods that could resist an attack from a future quantum computer that is more powerful than the comparatively limited machines available today. The selection constitutes the beginning of the finale of the agency’s post-quantum cryptography standardization project.” [\[23\]](#)

Although NIST has announced finalists, the research and standardization process will continue. Further testing, analysis, and potential modifications to the algorithms are planned before they are finally incorporated into the standards.

“For general encryption, used when we access secure websites, NIST has selected the CRYSTALS-Kyber algorithm. Among its advantages are comparatively small encryption keys that two parties can exchange easily, as well as its speed of operation.

For digital signatures, often used when we need to verify identities during a digital transaction or to sign a document remotely, NIST has selected the three algorithms CRYSTALS-Dilithium, FALCON and SPHINCS+ (read as “Sphincs plus”). Reviewers noted the high efficiency of the first two, and NIST recommends CRYSTALS-Dilithium as the primary algorithm, with FALCON for applications that need smaller signatures than Dilithium can provide. The third, SPHINCS+, is somewhat larger and slower than the other two, but it is valuable as a backup for one chief reason: It is based on a different math approach than all three of NIST’s other selections.” [\[23\]](#)

The announcement of the finalists in the NIST competition marked a breakthrough moment for post-quantum cryptography. The selected algorithms will provide the foundation for security in the era of quantum computing, replacing, or complementing current cryptographic methods. Organizations, governments, and businesses around the world are beginning to prepare to implement these new standards to ensure data is protected in the face of future technological threats.

2.3. Quantum computers

A subchapter in which the most notable events in the history of the creation and operation of quantum computers are presented, however, as in the case of quantum algorithms, the most valuable information on the method of operation and types of these machines are presented in the following chapters of this master's thesis. In particular, in chapter number 5.

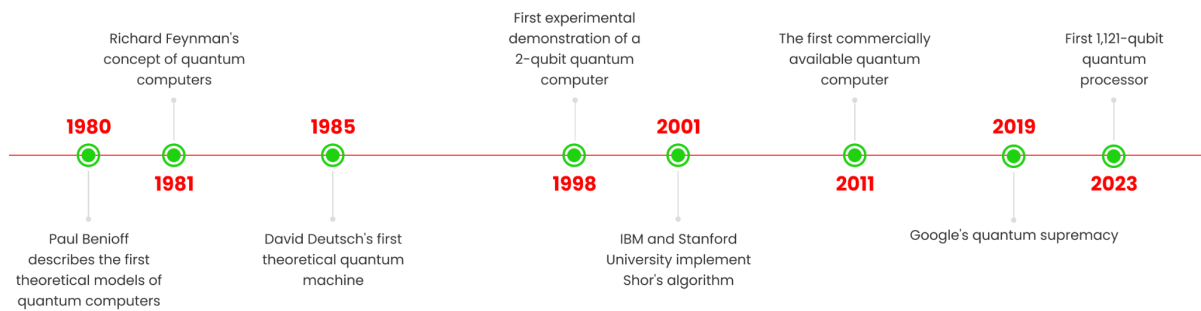


Figure 2.3: History of Quantum Computers (Created by the Author)

2.3.1. 1980: Paul Benioff described the first theoretical models of quantum computers. [\[13\]](#).

In 1980, Paul Benioff published a paper in which he presented the first theoretical model of a quantum computer, treating computations as quantum processes. His work showed that quantum mechanics could be used to model computations in a way that was consistent with quantum physics, a key step in the development of quantum computing theory. With this contribution, Benioff opened the door to further research in quantum computing, which was later built on by scientists such as Richard Feynman, David Deutsch, Peter Shor, and many others.

2.3.2. 1981: Richard Feynman proposed the concept of quantum computers [\[14\]](#).

During the **first conference on physics and computation** in MIT, he proposed the idea of using the principles of quantum mechanics to simulate complex quantum systems, something that classical computers could not do efficiently. Feynman suggested that a computer operating on quantum principles could perform such simulations much more efficiently, which became the foundation for the development of quantum computer theory. His work contributed to the understanding that quantum computers could potentially solve problems that were inaccessible to classical computers.

2.3.3. 1985: David Deutsch described a theoretical quantum machine [\[15\]](#).

In 1985, Deutsch published a paper in which he described a universal quantum computer that could execute any computational algorithm, introducing the foundations for the theory of quantum computation. Deutsch showed that a quantum computer, using the phenomena of superposition and entanglement, could simulate any physical process, including those that are impossible to efficiently simulate on classical computers. His work laid the theoretical foundations for the development of quantum computing and quantum algorithms.

2.3.4. 1998: First experimental demonstration of a two-qubit quantum computer is achieved using nuclear magnetic resonance (NMR) technology [\[16\]](#).

To properly understand this issue, at this stage it is needed to answer the question: What is a **qubit**? A **qubit** is the basic unit of information in quantum computers, which, unlike the classical bit that takes on a value of zero or one, can exist in a state of superposition, being simultaneously in the state of zero, one, and any combination thereof. This unique property, resulting from the principles of quantum mechanics, allows for the parallel processing of many possible states, which is the basis for the powerful computational capabilities of quantum computers.

In 1998, Isaac Chuang from Los Alamos National Laboratory, Neil Gershenfeld from MIT, and Mark Kubinec from the University of California at Berkeley built the first functional quantum computer, utilizing a two qubits system. Although their device-maintained coherence for only a few nanoseconds and was limited in solving practical problems, it successfully demonstrated the foundational principles of quantum computation.

If a quantum computer has a computing power of two qubits, meaning that it is able to simultaneously represent and process a superposition of four states (precisely $2^2 = 4$). Each additional qubit doubles the number of possible states the computer can process simultaneously, so a computer with two qubits can work with combinations of states: $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.

2.3.5. 2001: IBM and Stanford University run Shor's algorithm on a seven-qubit quantum computer [\[17\]](#).

The first-ever implementation of Shor's algorithm was achieved in 2001 by a team of researchers from IBM and Stanford University. It was a breakthrough in quantum computing, as it proved for the first time that Shor's algorithm could be implemented in practice on a real quantum computer.

To achieve this, the team used nuclear magnetic resonance (NMR) to create a seven-qubit quantum computer. In NMR, the qubits were represented by the nuclei of atoms in molecules, and their quantum states were controlled using radio pulses.

In the experiment, the researchers factored the number fifteen, which is simple in classical terms but was significant as a proof of concept. Shor's algorithm correctly factored the number fifteen into its prime factors, three and five.

2.3.6. 2011: D-Wave Systems announces first commercially available Quantum Computer [\[18\]](#).

Architecture:

Hundred and twenty-eight qubits chipset: D-Wave One was based on a Hundred and twenty-eight qubits chipset: called Rainier. Qubits (quantum bits) were implemented here as superconducting Josephson loops, which could represent both zero and one states simultaneously thanks to quantum entanglement and superposition.

Quantum Annealing:

D-Wave One was not a classical quantum computer based on the quantum gate model, like later IBM or Google computers. Instead, it used quantum annealing. This is an optimization method that allows finding the global minimum of a function by exploiting quantum effects. Quantum annealing is particularly useful for solving optimization problems, such as finding the shortest path in a graph or minimizing energy in physical systems.

2.3.7. 2019: Google claims to achieve "quantum supremacy" with its Sycamore quantum processor [\[19\]](#).

In the experiment, Sycamore solved a problem related to randomly sampling quantum circuits in two hundred seconds. According to Google, the fastest classical supercomputer would need thousands of years to do this task.

This meant that for the first time a quantum computer had performed a task that is impossible for traditional computers to perform in a reasonable time. This was a crucial step in the development of quantum computers, but experts point out that this achievement concerned a specific problem and does not mean that quantum computers are already capable of solving a wide range of practical problems.

However, at this stage, to approach the subject fairly, it is necessary to present the other side of the coin, namely the criticism that Google received from its competitors.

The speed experiment developed by Google faced criticism from experts and competitor companies after its results were leaked online prior to the official release. IBM responded in a blog post Monday, arguing that Google had overstated the complexity of the computing task. According to IBM, a classical computer could solve the problem in just two and a half days, rather than the ten thousand years suggested by Google.

“We urge the community to treat claims that, for the first time, a quantum computer did something that a classical computer cannot with a large dose of skepticism,” ~ said an IBM representative [\[67\]](#).

2.3.8. 2023: IBM unveils the IBM Quantum Condor, the first 1,121-qubit quantum processor [\[20\]](#).

The IBM Condor, with one thousand one hundred twenty-one qubits, is one of the latest and most advanced quantum processors developed by IBM. The processor represents a key step in the development of quantum computers and aims to overcome the limitations of current technologies. The IBM Condor is designed to scale quantum systems, meaning it can manage significantly more qubits while reducing errors, which is key to practical quantum applications.

Condor is expected to be a turning point towards achieving "quantum advantage."

Its introduction is also a step towards building scalable quantum systems of the future that will be able to integrate with other quantum processors, allowing for further increases in the computing power of these machines.

IBM plans to continue developing and scaling its quantum processors, with additional devices planned to exceed one million qubits in the coming years.

This chapter has demonstrated the noteworthy progress required in the field of quantum mechanics, as well as in quantum and post-quantum algorithms and computers, to enable the implementations and experiments presented in this thesis. The outline of this chronology provides a foundation for further considerations in the thesis, showing the context and motivation for research on cryptography resistant to quantum attacks. If even one of the above-mentioned events had not happened, this master's thesis might never have been written.

3. Quantum computing - A theoretical approach to explaining quantum computing issues.

A chapter whose role is to explain the basic issues and concepts related to quantum mechanics. To introduce the most important discoveries, research, and their results. Provide the reader with the necessary theoretical knowledge to understand further parts of the work. Key concepts such as superposition and quantum entanglement are introduced here, which form the foundation for the analysis of quantum and post-quantum algorithms. With this knowledge, even a reader without specialized background in quantum physics will be able to properly interpret and evaluate the issues and research results discussed in the paper. It serves as an introduction to the sections that follow, discussing quantum algorithms, quantum computers and how they work, while drawing directly from the chapter on the history of quantum mechanics, developing the topics covered there.

3.1. Fundamentals of Quantum Mechanics

Quantum mechanics is a physical theory that develops classical mechanics, making it possible to properly describe the micro-world, that is, individual elementary particles and their systems, such as atoms or ions [\[24\]](#).

A characteristic of quantum mechanics is that it does not directly describe the variability of measurable physical quantities, such as the position or velocity of objects. Instead, it deals with the description of an abstract state vector and its influence on the results of measurements, which are partially random, that is indeterministic in nature. This leads to paradoxical phenomena such as:

3.1.1. Quantization of physical quantities

In quantum mechanics, physical quantities such as energy or angular momentum can only take on discrete values, which contrasts with classical mechanics, where these quantities are continuous.

Quantization is a procedure that enables the transition from classical field theory to quantum field theory. It involves replacing classical variables such as position and momentum, described by time- and space-dependent functions, with quantum operators. The process is a generalization of the method used in the transition from classical mechanics to quantum mechanics.

It is also worth emphasizing at this stage that the quantum field theories discussed here are not a direct component of quantum mechanics, but only a development of it that ensures compatibility with the special theory of relativity. The combination of these two concepts forms **quantum physics**.

In collective understanding, quantization refers to the existence of a finite or countable set of permissible values of a given quantity. For example, to say that the energy of an electron in an atom is quantized meaning that it can only take on certain values, called energy levels.

3.1.2. Wave–particle duality of Louis de Broglie [\[5\]](#):

It is a property of quantum objects, such as photons or electrons, that they can exhibit wave-like (for example diffraction and interference) or particle-like (for instance precise position and momentum) characteristics, depending on the circumstances.

According to the theory of quantum mechanics, it is not only quantum objects, but also all matter that exhibits such duality, although it is directly revealed only in very subtle experiments to be performed on small particles such as atoms, photons, and other quantum objects.

Double-slit experiment

In a classical double-slit experiment, photons exhibit interference when they are not observed, demonstrating their wave nature. However, when detectors are placed at the slits to record which slit a photon passes through, the interference disappears, and the photons begin to behave like particles. This illustrates how the act of measurement affects the state of the photon, according to the indeterminacy principle. This is an experiment that proves the corpuscular-wave nature of photons.

A pictorial representation of this experiment can be observed below.

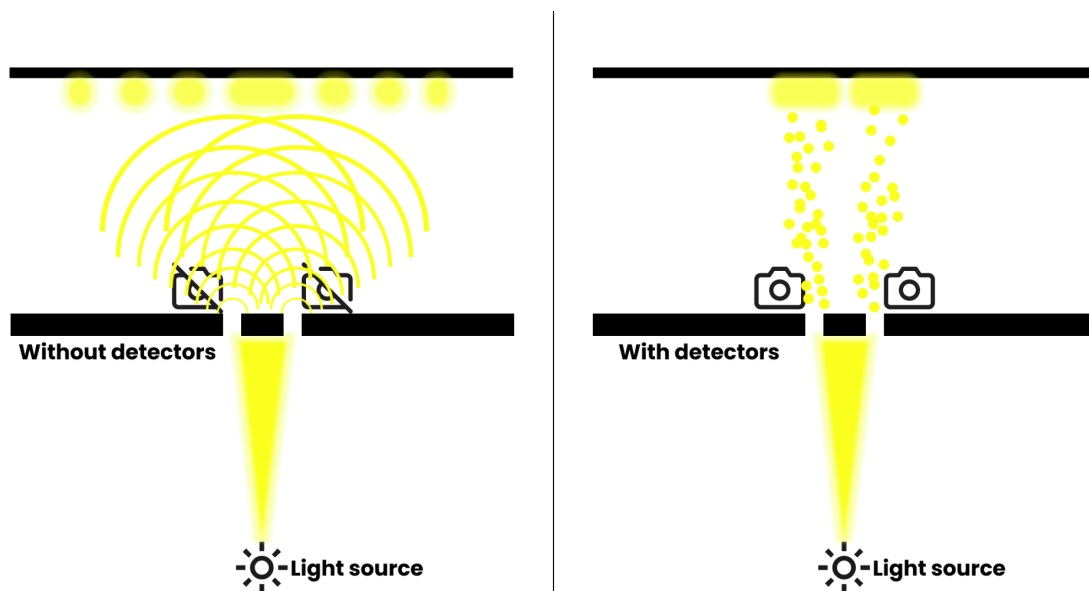


Figure 3.1: Double-slit experiment (Created by the Author)

The above experiment (Figure 3.1), which is, among other things, proof of the wave properties of light, is the “**Young's Experiment**,” which was first performed as early as 1801. Its visible result is the so-called interference striations. Below, in the own-elaborated graphic the reasons for such a phenomenon can be observed.

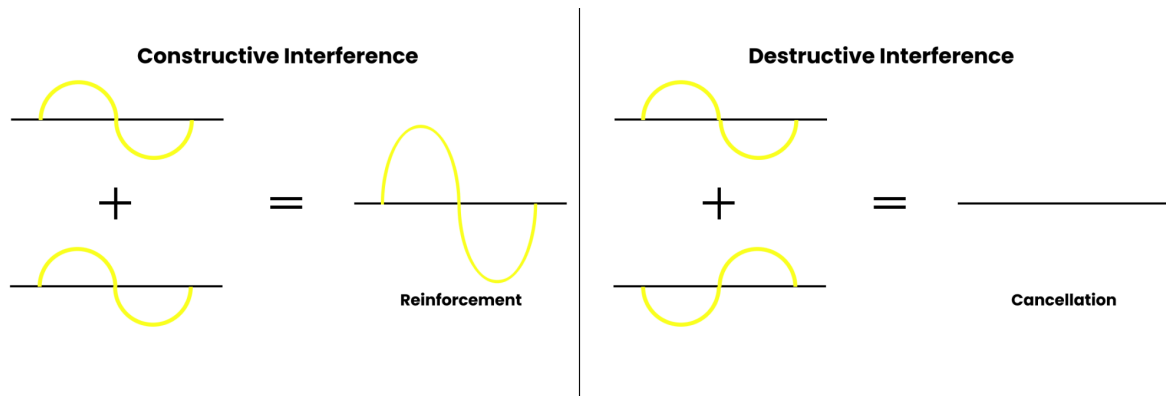


Figure 3.2: Interference (Created by the Author)

Interference is a phenomenon that occurs when two waves overlap. In the case of light waves, interference can lead to amplification (constructive interference) or attenuation (destructive interference) of light intensity at different points in space.

Constructive interference occurs when the waves meet in phase, meaning that the ridges of one wave coincide with the ridges of the other. In this case, the amplitudes of the waves add up, leading to signal amplification (greater brightness).

Destructive interference occurs when waves meet in counter-phase, meaning that the crests of one wave coincide with the valleys of the other. In this case, the amplitudes of the waves subtract, leading to a weakened signal (less brightness, in extreme cases - total darkness).

Given the importance of this issue in later chapters, particularly the fifth, it is worth noting at this stage that, as with the interference of light waves discussed above (Figure 3.2), the subject of quantum computers also involves the phenomenon of interference - it refers to one of the key phenomena of quantum mechanics that are used to perform calculations. It is a process in which the quantum waves associated with different states of qubits overlap, leading to an amplification of some probabilities of results and a weakening of others. In the context of quantum computing, interference is used to constructively amplify the probability of getting the correct solution to a problem, while eliminating incorrect answers.

3.1.3. Heisenberg's indetermination principle, quantum superposition, wave function collapse, Schrödinger's cat [\[8\]\[25\]\[26\]](#).

Intrinsically linked concepts that, for good understanding, are worth quoting and discussing in one place.

The indeterminacy principle, formulated by Werner Heisenberg, states that certain pairs of physical quantities, such as the position and momentum of a particle, cannot be measured accurately at the same time with arbitrarily high precision. The act of measurement affects the quantum state of the photon. This principle is fundamental to quantum mechanics because it illustrates that there are limits to our ability to know the exact values of a particle's physical quantities, because of its wave nature.

Quantum superposition means that particles can be in multiple states simultaneously until a measurement is made that forces a “**collapse**” of the superposition state to one of the possible states. A good illustrative example to help understand this topic is again, shown above (Figure 3.1), the “**Double-Slit Experiment**.”

Measurement problem (**wave function collapse**):

Quantum mechanics describes the evolution of quantum states in a deterministic way using Schrödinger's equation, but the act of measurement itself causes a “**collapse**” of the wave function, which is an indeterministic and random process. This leads to questions about the nature of reality - whether the quantum state before measurement is “real” or exists only potentially.

Prediction vs. measurement: explaining how measurement in quantum mechanics affects the state of the system, leading to the so-called “wave function collapse.”

In addition to the experiment with a double slit, discussed in connection with the corpuscular-wave nature of light, there is another incredibly famous thought experiment that illustrates the paradoxes of quantum superposition and measurement.

It is, of course, the “**Schrödinger's Cat**” [\[26\]](#).

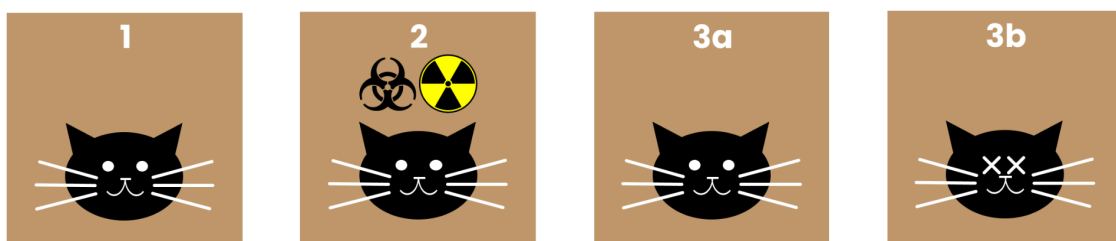


Figure 3.3: Schrodinger's Cat (Created by the Author)

Step-by-step experiment:

1. A live cat is placed in the container.
2. A single atom of a radioactive element, a poisonous gas, and a sensor capable of detecting whether our atom has emitted a particle of ionizing radiation are still inserted into the box. If so, the mechanism releases the poisonous gas. The container is sealed tightly.

What is important to emphasize and understand the unpredictability of this experience: Radioactive decay is when the atomic nucleus of a radioactive element spontaneously and unpredictably emits particles, moving to a lower energy state. Each individual nucleus has a certain probability of decaying at a given time, but there is no way to predict exactly when a given atom will decay. The decay is therefore a quantum phenomenon, characterized by fundamental randomness.

3. Observing the closed box, common sense would indicate that there is a fifty percent chance that the cat is still alive, or that it is already dead. However, quantum physics comes with a different explanation for this phenomenon. Making the cat's fate dependent on just one atom brings it under the power of quantum physics. According to its principles, Schrödinger's cat is in **superposition**, that is, until the box is opened, it remains both alive (3a) and dead (3b) at the same time (Figure 3.3). Only by opening the box does the **wave function collapse**, already described above, and the cat unequivocally assumes one state.

An important concept in the context of this thought experiment is **Decoherence**.

In the context of Schrödinger's cat experiment, it explains why it is impossible to observe superposition of states on a macroscopic scale (a cat simultaneously alive and dead). Decoherence is a process in which a quantum superposition becomes “blurred” due to **quantum noise** resulting from interactions with the environment, causing the quantum system to stop behaving like a superposition and “collapse” into one of the classical states (a living or dead cat). This is because, on a macroscopic scale, interactions with the environment are inevitable. That is why a cat is alive or dead, and not in both states at the same time.

Quantum noises are unwanted fluctuations or disturbances in a quantum system that can lead to errors in quantum calculations. They arise for a variety of reasons, such as interactions with the environment, imperfections in the apparatus, or natural energy fluctuations at the quantum level. These noises can disturb the delicate quantum states of qubits, causing errors in logic operations and decoherence.

Is it then possible to replicate this thought experiment on a micro-scale? In a perfectly preserved environment in which the decoherence phenomenon does not occur. It does.

“Schrödinger cat states of a sixteen-microgram mechanical oscillator” [\[27\]\[28\]\[29\]](#).

In April 2023, researchers at ETH Zurich conducted an experiment that is a modern, and most importantly physical, not just a thought version of Schrödinger's famous cat experiment, creating a “Schrödinger cat state” for a mechanical oscillator. This is the heaviest object so far in such a quantum state, which means that this object, which is a crystal with a mass of sixteen micrograms, can be in a superposition of two states of oscillation at the same time - this can be compared to a situation where Schrödinger's cat is both alive and dead at the same time.

In this experiment, the crystal oscillates simultaneously in two opposite directions, which is the quantum equivalent of the cat's “living” and “dead” states. Although the mass of this object is much larger than that of atoms or molecules, it is still several billion times smaller than that of a real cat, but this makes this achievement unique. The researchers also observed decoherence, which will help better understand why quantum effects disappear in the macroscopic world.

3.1.4. Spin and quantum entanglement [\[30\]](#).

Spin and entanglement are closely related in quantum mechanics and together play a key role in many quantum phenomena, including quantum teleportation and quantum cryptography.

The spin of elementary particles is their intrinsic quantum property, which can be compared to angular momentum. However, it is quite different from the classical concept of spin. Spin values are quantized and can take on specific values, such as $\frac{1}{2}$, 1, 0. Spin is important for many quantum phenomena, including the behavior of particles in a magnetic field.

Quantum entanglement:

Quantum entanglement is a phenomenon in which two or more particles become so interconnected that the state of one particle is related to the state of the other, regardless of the distance that separates them. When particles are entangled, measuring the spin of one particle immediately determines the spin state of the other, even if they are a great distance apart.

This phenomenon, which Albert Einstein called “spooky action at a distance,” contradicts classical notions of locality, where information cannot travel faster than light.

Entanglement requires particles to be in a pure quantum state, or superposition of states. Any interaction with the environment that introduces decoherence (for example thermal perturbations, interactions with other particles) can destroy entanglement.

Spin vs entanglement:

Entanglement most often involves the spin of particles. For example, if two electrons are entangled, their spins are related to each other in such a way that if one electron has an “up” spin, the other must have a “down” spin, and vice versa. If no measurement is made, the two spins are in superposition, but the measurement of one immediately determines the spin of the other.

Spin entanglement is central to many quantum experiments and applications, as it enables the instantaneous linking of particle states over long distances, which has no equivalent in classical physics. This phenomenon is one of the most fascinating aspects of quantum mechanics and has been a key element in discussions about the nature of quantum reality, including the famous **EPR (Einstein-Podolsky-Rosen)** paradox, which has already been addressed and explained in this work at the stage of presenting the history of quantum mechanics.

3.1.5. Quantum fluctuations [\[31\]](#).

Quantum fluctuations are spontaneous, instantaneous changes in physical values, such as energy, electromagnetic field, or position of a particle, which occur even in a vacuum. They arise from Heisenberg's indeterminacy principle, which states that a pair of quantities (such as energy and time) cannot be determined simultaneously with any precision.

Significance: These fluctuations are fundamental to the operation of qubits, which can exist in superposition states, enabling the parallel processing capabilities that give quantum computers their potential superiority over classical computers. However, quantum fluctuations also introduce challenges, such as noise and decoherence, which can affect the stability and reliability of quantum algorithms. Therefore, understanding and managing quantum fluctuations is essential for the development and optimization of quantum technologies.

3.1.6. Quantum jumps [\[32\]](#).

Quantum jumps are sudden changes in the quantum state of a particle, usually associated with a transition between discrete energy levels. They occur, for example, when an electron in an atom jumps from one energy level to another, **emitting or absorbing a quantum of energy (photon)**.

Significance: Quantum jumps are central to understanding the processes of light emission and absorption, as well as to the operation of many quantum devices, such as lasers and quantum computers.

3.1.7. Quantum emission and absorption [33].

Quantum emission and absorption are processes in which a particle (usually an electron in an atom) transitions between discrete energy levels, emitting or absorbing a quantum of energy in the form of a photon.

- **Quantum emission:** It occurs when an electron moves from a higher energy level to a lower one, emitting a photon with an energy equal to the difference between these levels.
- **Quantum absorption:** This occurs when an electron absorbs a photon and moves from a lower energy level to a higher one.

3.1.8. Quantum tunneling [34]:

Quantum tunneling is a phenomenon in which a particle passes through a potential barrier, even though classically it would not have enough energy to do so. In classical mechanics, the particle would have to have an energy equal to or greater than the barrier, but in quantum mechanics, due to its wave nature, there is some probability that the particle will “jump” over the barrier. Tunneling is crucial to many physical processes, such as nuclear reactions in stars, the operation of semiconductor junctions in electronics, and scanning tunneling microscopes (STMs). For example, an electron can tunnel through an insulator in a tunnel junction, a key phenomenon in semiconductor technology.

3.1.9. Quantum operators [35].

Quantum operators are mathematical objects that in quantum mechanics correspond to measurable physical quantities, such as energy, momentum, or position. They act on wave functions (quantum states) and transform them into new wave functions.

For example, the momentum operator acts on the wave function to give information about the momentum of a particle in that state. These operators are key in quantum equations such as the Schrödinger equation, where the Hamiltonian operator (describing energy) allows for finding **eigenstates and eigenvalues** of the system's energy.

3.1.10. Eigenstates & eigenvalues

In quantum mechanics, **an eigenstate** is a quantum state that remains unchanged (except for a possible change in scale) after a certain operator is applied to it, such as the Hamiltonian operator that describes the energy of a system. If a particle is in the eigenstate of a given operator, a measurement of the corresponding physical quantity (e.g., energy) will always yield a certain value. **An eigenvalue** is a number that results from the action of an operator on its eigenstate. In the context of quantum mechanics, an eigenvalue corresponds to a measurable physical value, such as energy, when the system is in a given eigenstate.

4. Quantum Algorithms - Presentation and description of the most popular & groundbreaking algorithms of quantum cryptography

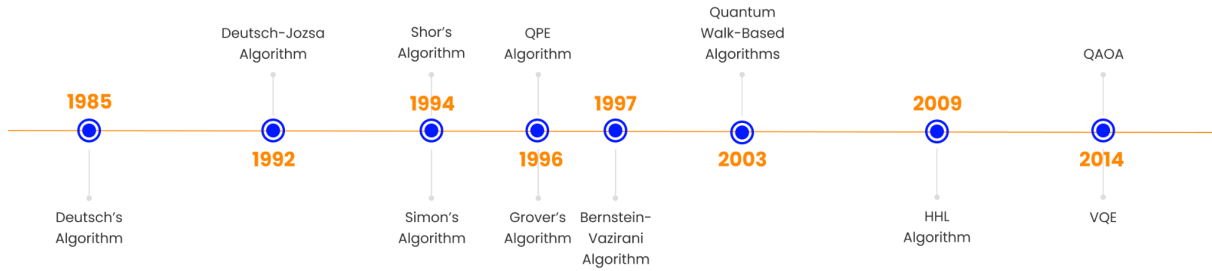


Figure 4.1: Extended history of Quantum Algorithms (Created by the Author)

This chapter is devoted to the most important and groundbreaking quantum algorithms. It discusses the unique (unique) purpose of each of them, their impact on the development of the present field and their mode of operation. However, the following chapter will lack the algorithms considered the most important, namely Shor and Grover, as they have been given their own chapters in this dissertation, in which they will be discussed, implemented, and evaluated in detail.

Deutsch's Algorithm (1985) [\[36\]](#).

Significance: First quantum algorithm demonstrating that quantum computers can solve certain problems faster than classical computers, even if the problem is simple.

The Deutsch algorithm is one of the basic quantum algorithms, proposed by David Deutsch in 1985. Its purpose is to solve a problem in which it is needed to determine whether a given function $f(x)$ is constant (means that it always returns the same value for each input) or differential (means that it returns different values for at least two different inputs).

In the classical case, the function $f(x)$ must be called at least twice to determine this. However, thanks to the superposition of states in a quantum computer, Deutsch's algorithm can do this with only one function call, using **quantum gates** and the interference principle.

Quantum gates are basic operations in quantum computing, like logic gates in classical computers. However, instead of operating on bits that can take the value zero or one, quantum gates operate on qubits (quantum bits) that can be in state zero, one or any superposition of these states. Quantum gates are usually represented as matrices.

Deutsch-Jozsa Algorithm (1992) [\[37\]](#).

Inventors: David Deutsch and Richard Jozsa

Significance: An improvement over Deutsch's algorithm, it solves a specific problem exponentially faster than any deterministic classical algorithm.

The Deutsch-Jozsa algorithm extends the problem presented above to a function $f(x)$ that takes an n -bit string (instead of a single bit) and returns a value of zero or one. This function can be:

- Constant: Returns the same value (zero or one) for every possible n -bit input.
- Balanced: Returns zero for exactly half of the n -bit inputs and one for the other half.

In the classical approach, to distinguish whether a function is constant or balanced, it is needed to check many inputs (in the worst case, $2^{n-1} + 1$ trials).

The fundamental change is that the Deutsch-Jozsa algorithm, thanks to the use of superposition and interference in quantum computing, can make the same classification (whether the function is constant or balanced) in one step - regardless of the number of bits n in the input.

In summary, the main difference between the Deutsch algorithm and the Deutsch-Jozsa algorithm is the scale of the problem they solve (from one bit to an n -bit string), and the fact that the Deutsch-Jozsa algorithm shows an even more pronounced quantum advantage compared to the classical approach with larger inputs.

Simon's Algorithm (1994) [\[38\]](#)[\[39\]](#)[\[40\]](#).

Inventor: Daniel Simon

Significance: Solves the "Simon's problem," showing an exponential separation between the query complexities of quantum and classical computers for a specific problem.

Daniel Simon's algorithm is one of the first quantum algorithms that showed that quantum computers can be much faster than classical ones in solving certain problems. It was proposed in 1994 and is a crucial step in the development of the theory of quantum computing. Served as the inspiration for Shor's algorithm.

A function $f(x)$ that takes an n -bit string (for example, xxx can be 000, 001, ..., 111) and returns another n -bit string. This function has a special property: there is a mysterious n -bit string s that makes it so that if two different inputs x and y give the same result $f(x) = f(y)$, the difference between x and y is exactly s (expressed as an XOR operation). The task of this algorithm is to discover this mysterious string s .

To find this string using a classical computer, it would have to make an enormous number of attempts - about $2^{n/2}$ calls of the function $f(x)$. This quickly becomes infeasible when the number of bits n is large.

Simon's algorithm takes advantage of quantum phenomena such as superposition and interference, allowing it to find the string s much faster (in linear time, whereas a classical computer would take exponentially more time to do so) - using only about n calls to the function $f(x)$.

Quantum Phase Estimation Algorithm (QPE) (1996) [\[41\]](#).

Inventors: Multiple contributors, often attributed to Kitaev and others

Significance: A fundamental algorithm used in many other quantum algorithms, including Shor's algorithm, for estimating the phase (eigenvalue) of an eigenvector of a unitary operator.

Quantum Phase Estimation (QPE) Algorithm is a key component of many quantum algorithms, such as Shor's algorithm (for number factorization) and quantum chemistry algorithms. Its main purpose is to estimate the eigenvalues of a **unitarity matrix**, which leaves the **eigenvector** unchanged.

Unitary matrix is a special type of matrix that is commonly used in quantum computing. A unitarian matrix has the property that its eigenvalues lie on a unit circle in the complex plane.

An eigenvector is a special vector that, when multiplied by a certain matrix (in this case, a unitarian matrix U), does not change direction, but can change length (can be multiplied by an eigenvalue).

QPE is based on the idea that with the help of quantum computing it is possible to find the phase associated with the eigenvector of a given unitarity matrix. In classical computers, this type of calculation would be very time-consuming, but quantum computers can do it efficiently thanks to superposition and qubit entanglement.

Two main approaches:

1. Kitaev's approach

In this version of QPE, the phase is determined using Hadamard tests, and then classical calculations are performed to accurately determine the phase. This approach requires fewer quantum operations but requires classical processing after the quantum calculations are performed.

2. Approach with Quantum Fourier Transform

In this approach, the phase is determined using the quantum Fourier transform. This approach is more intuitive from a quantum computing point of view, but it requires precise rotation operations, which can lead to larger reading errors on actual quantum computers.

However, the implementation of QPE is quite challenging, as it faces difficulties on current quantum computers, such as those within the Noisy Intermediate-Scale Quantum (NISQ) technology, due to the low accuracy and high variability of the physical properties of these computers. To obtain accurate results, it is necessary to minimize the number of controlled rotation operations, which limits the depth of quantum circuits and reduces measurement errors.

QPE is a powerful tool in quantum computing to quickly find eigenvalues of matrices, which is crucial in many applications such as factorization, quantum chemistry and machine learning algorithms. However, its implementation on real quantum computers requires precision and optimization to get accurate results.

Bernstein-Vazirani Algorithm (1997) [\[42\]](#).

Inventors: Ethan Bernstein and Umesh Vazirani

Significance: This algorithm was developed as an improvement over Deutsch's algorithm, and it demonstrated how a quantum computer could solve a problem more efficiently than a classical computer. The algorithm determines a hidden string of bits with a single query to the quantum oracle, whereas a classical computer would require multiple queries. Although it does not offer an exponential speedup, it is significant as one of the early quantum algorithms showing the potential for quantum advantage.

It is used to solve the problem of discovering some hidden information in Boolean functions in a much faster way than classical algorithms.

The premise of the problem is this: a black box (Oracle function - which computes $f(x)$ for every possible input x , which introduces the appropriate phases into the superposition) that represents some boolean function $f(x)$ exists. This function is linear, which means that it can be expressed as $f(x) = a \cdot x \pmod{2}$, where:

- x is the n -bit input vector,
- a is an n -bit vector to be discovered,
- “ $\cdot$ ” means scalar product (i.e., the sum of the products of the bits modulo 2).

The Bernstein-Vazirani algorithm is designed to discover the vector a that defines the function $f(x)$.

The Bernstein-Vazirani algorithm is particularly efficient because in the classical case, discovering the vector a would require 2^n trials in the worst case, while the BV algorithm discovers a with just one Oracle function call and a few quantum operations.

The BV algorithm, while of theoretical importance as one of the first demonstrations of the superiority of quantum algorithms over classical ones, also has practical applications in cryptography, especially

in the analysis of Boolean functions used in the construction of ciphers and attacks on them. It is also the basis for understanding more advanced quantum algorithms, such as Grover's and Shor's algorithm.

Quantum Walk Algorithms (around 2003) [\[43\]](#).

Inventors: Various, with significant contributions by Andris Ambainis and others

Significance: Quantum walks generalize classical random walks and can be used to speed up certain types of algorithms, such as those for element distinctness or search on graphs.

Quantum search algorithms, with Grover's algorithm at the forefront, are among the most important developments in quantum computing. Their main goal is to find a labeled element in a large, unstructured data set faster than is possible with classical algorithms.

The classical linear search algorithm requires, in the worst case, searching every element in a set of size N , which means $O(N)$ operations. Quantum search algorithms, particularly Grover's algorithm, significantly speed up this process, reducing the number of required operations to $O(\sqrt{N})$. As a result, Grover's algorithm and other quantum search algorithms are extremely efficient in searching large data sets. They have applications in cryptography (e.g., password cracking) and other fields that require fast searching of huge state spaces.

These algorithms not only demonstrate the advantages of quantum computing over classical computing in the context of searching unstructured sets, but also provide a basis for further research and development of more advanced quantum techniques.

HHL Algorithm (2009) [\[44\]](#).

Inventors: Aram Harrow, Avinatan Hassidim, and Seth Lloyd

Significance: The Harrow-Hassidim-Lloyd (HHL) algorithm solves systems of linear equations exponentially faster than the best-known classical algorithms under certain conditions.

The HHL algorithm (Harrow, Hassidim, Lloyd) proposed in 2009 is one of the key quantum algorithms for solving problems involving systems of linear equations on quantum computers. Although it is closely related to the classical solution of systems of linear equations ($f(x) = a * x$), the HHL algorithm solves a specific quantum linear equation system problem (Quantum Linear System Problem, QLSP).

Quantum Speed: The HHL algorithm offers the quantum advantage of being able to solve systems of linear equations in logarithmic time relative to the size of the matrix, while classical methods require linear or quadratic time depending on the algorithm used.

Implementation: The HHL algorithm is based on several key quantum techniques, such as quantum Fourier transform, techniques related to homogeneous operators (Hamiltonian simulation), and phase measurement (phase estimation). These advanced operations allow solutions to be extracted from the quantum state.

The HHL algorithm is the foundation for many modern quantum applications and is one of the most promising tools for solving problems that classical computers solve much more slowly or not at all. Its impact on the development of quantum machine learning techniques and other advanced fields cannot be overestimated, although it also entails some technical challenges that are still being studied.

Variational Quantum Eigensolver (VQE) (2014) [\[45\]](#).

Inventors: first developed by Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O'Brien

Significance: A hybrid quantum-classical algorithm used to approximate the ground state energy of a quantum system, particularly useful in quantum chemistry.

Variational Quantum Eigensolver (VQE) is one of the most promising quantum algorithms, especially in the context of the Noisy Intermediate-Scale Quantum (NISQ) era of quantum computers. It was developed by Peruzzo with his team in 2014, and its theoretical basis was developed in the work of McClean with his team.

VQE is a hybrid algorithm that combines quantum and classical calculations to find the ground state of a **quantum Hamiltonian**, and thus its lowest **eigenvalue**, which is crucial in fields such as quantum chemistry and condensed matter physics. **Eigensolver** is a general term describing such an algorithm, finding eigenvalues (eigenvalues) and eigenstates (eigenstates) of quantum systems.

The quantum Hamiltonian is an operator in quantum mechanics that represents the total energy of a quantum system. It is a central tool in describing the dynamics and behavior of quantum systems. For example, it is used in the Schrödinger equation, which describes the time evolution of a quantum state.

VQE has broad applications in studying the electron structure of molecules, which is crucial in quantum chemistry and materials engineering. Potential applications include drug discovery, the design of new materials, and other fields that require accurate calculations on large quantum systems. It is also the basis for many quantum machine learning and simulation algorithms because it can model complex quantum interactions that are difficult to compute on classical computers, especially as systems become larger.

Quantum Approximate Optimization Algorithm (QAOA) (2014) [\[46\]](#)

Inventors: Edward Farhi, Jeffrey Goldstone, and Sam Gutmann

Significance: A quantum algorithm designed to solve combinatorial optimization problems by approximating the solution to certain optimization problems.

QAOA (Quantum Approximate Optimization Algorithm) is a hybrid quantum-classical algorithm designed to solve combinatorial optimization problems. It is particularly promising in the context of quantum computers known as NISQ (Noisy Intermediate-Scale Quantum) devices, which have a limited number of quantum operations due to noise and decoherence.

QAOA is used in situations where it is necessary to find the best solution (e.g., minimize cost or maximize profit) among many possible combinations, a task that is difficult to solve using classical computational methods, especially when the size of the problem increases.

The algorithm has potential applications in many fields, such as logistics, production planning, network optimization, and other problems where computational efficiency is key. It can also be used to solve optimization problems in quantum chemistry, physics, and finance.

Although QAOA and VQE are different algorithms, they share common features:

- Both are hybrid quantum-classical algorithms.
- Both use parameter optimization on a classical computer.
- Both algorithms prepare a quantum state, which is then evaluated and optimized to obtain the desired result.

QAOA is specifically designed for combinatorial optimization, while VQE is more related to problems of quantum chemistry and finding eigenstates of the Hamiltonian.

5. Quantum Computers - Introducing the subject of quantum computers (especially IBM's) and how they work.

This chapter was created in the form of answers to the most important and intriguing questions in the field of quantum computers [\[20\]\[47\]](#). This is a particularly accessible figure to introduce the reader to the topic in an interesting way, omitting the historical aspects that have already been discussed in subchapter 2.3, as well as without explaining the most important concepts to which the entire third chapter of this scientific work is devoted.

5.1.1. What are quantum computers, what are their main operating principles, and how do they differ from classical quantum computers?

Quantum computers are computational devices that use the principles of quantum mechanics, such as superposition, entanglement, and interference, which are used to manipulate information in ways that classical computers cannot. The basic unit of information in a quantum computer is a qubit, which, unlike the bit that classical computers operate on, can be in the state of zero, one, or their superposition (more on this is said in subsection 2.3.4, when discussing the first quantum computer, with the power of two-qubits). Quantum computers can process a vast number of states simultaneously, giving them the potential to solve problems that are inefficient or impossible for classical computers to solve. They also differ in the way they process information, where **classical logic gates** are replaced by **quantum gates**. Complex structures consisting of a sequence of quantum gates are **quantum circuits**.

	Classical logic gates	Quantum gates
Operation type	They operate on bits (0 or 1)	They operate on qubits (0, 1, superposition)
Basic function	AND, OR, NOT, XOR	Hadamard, Pauli-X, CNOT, phase gates
Determinism	Deterministic results of operations	Results dependent on quantum states (superposition)
Reversibility	Usually, irreversible	Reversible
Operation complexity	Simple, unambiguous logical operations	Ability to perform complex multi-state operations
Use of interference	Does not use	Uses to manipulate states

5.1.2. How are they categorized? For what is each of them used?

Quantum computers can be classified based on the technology they are based on, such as photonic or gate systems. Photonic systems use photons to store and process quantum information, which is particularly useful in quantum communication. Gate systems, on the other hand, operate on qubits in a controlled manner using quantum gates, which makes them more versatile and capable of implementing a variety of quantum algorithms.

	Photonic quantum computers	Gate quantum computers
Information carrier	Photons	Superconducting circuits, ion traps
Principle of operation	Using the properties of photons (polarization, time of flight)	Logical operations on qubits using quantum gates
Applications	Quantum cryptography thanks to QKD, quantum communication, quantum simulations, specialized calculations (for example optical transformations)	General quantum calculations, optimization, accurate chemical modeling, Data analysis
Information transfer	Easy integration with quantum networks	More difficult integration with quantum communication
Advantages	Stability - Higher resistance to decoherence	Versatility - the ability to implement a wide range of quantum algorithms, such as Shor's or Grover's algorithm.
	Quantum communication - the ability to easily transmit photons over long distances,	Circuit shaping - the ability to program any quantum operations through appropriate gate configurations.

The IBM quantum computers on which Shor and Grover's algorithms are implemented in this work use gate technology. IBM uses superconducting qubits, which are implemented as logic gates operating on qubits within the gate architecture.

Superconducting quantum computers (IBM, Google), ion traps (for instance IonQ), and photon systems (for example Xanadu). Each of these has its own unique properties, challenges, and applications.

5.1.3. What are the different approaches in building quantum computers (for instance superconductor qubits, ion trap qubits, silicon qubits)? What are the differences between the technologies used?

There are many approaches to realizing qubits, such as:

- **Superconductor qubits**, which use superconducting current loops,
- **Qubits from ion traps**, where single ions are held in place by electromagnetic fields,
- **Silicon qubits**, based on electrons in silicon nanostructures.

Each of these approaches has its own unique advantages and challenges, such as qubit stability, ease of manipulation, and technology scalability. Different qubit technologies differ in terms of stability, coherence times, scalability, and ease of integration with existing technology infrastructure. For example, superconducting qubits have shorter coherence times but are easier to control at low temperatures, while ion trap qubits can have longer coherence times but are more difficult to scale.

5.1.4. What are the technological challenges involved in building quantum computers?

Key technological challenges include the problem of decoherence, that is. loss of quantum state of qubits due to interactions with the environment, and quantum noise, which can interfere with computations. In addition, the scalability of quantum systems and the precise control and reading of qubit states are significant obstacles to the development of practical quantum computers. The concepts of decoherence and quantum noise have already been introduced and explained in Section 3.1.3.

5.1.5. How to access a quantum computer? What languages to use?

The way to gain remote access to quantum computers is through quantum services in the cloud - Cloud Platforms. They enable remote access to quantum computers, which allows users to run quantum algorithms without the need for their own hardware. Examples include **IBM Quantum Experience**, Google Quantum AI, and Amazon Braket.

Once users have access, it is needed to know how to talk to such a computer, and the following specialized languages and tools for writing quantum algorithms come in handy: **Qiskit (IBM)**, Cirq (Google), or PyQuil (Rigetti).

5.1.6. How do quantum computers impact modern cryptographic methods?

The research results indicate that modern quantum computers and the algorithms used in them, at least at the current stage of development, do not pose a real threat to current cryptographic methods, including the RSA algorithm. Although quantum computers theoretically have the potential to compromise the security of some encryption methods, especially those based on integer factorization, such as RSA, the practical application of Shor's algorithm, which could effectively solve the factorization problem, is still in the experimental phase and does not threaten currently widely used cryptographic systems. Quantum computers and algorithms do not directly threaten modern cryptographic methods. However, they indicate the way for the entire field, demonstrating dynamic development and signaling potential threats to currently used cryptography in the future.

5.1.7. What are the risks and opportunities associated with the use of quantum computers in cryptography?

Quantum cryptography is a field of science that uses the principles of quantum mechanics, such as the uncertainty principle and quantum entanglement, to provide secure communication. One major example is quantum key distribution (QKD), which allows for the secure transmission of cryptographic keys in a way that is resistant to eavesdropping. The threats associated with quantum computers include the possibility of breaking current cryptographic systems, but the development of post-quantum cryptography and quantum cryptography offers new methods for securing data. Opportunities also include improving security through new methods of encryption and authentication.

5.1.8. What are the current capabilities of modern quantum computers, what are their limitations, and what are the prospects for the future?

Modern quantum computers can perform limited tasks, such as quantum simulations of simple chemical and physical systems, and experimental tests of quantum algorithms. However, their full potential is still in development, and most current systems are still susceptible to errors, such as **qubit decoherence**, which leads to loss of information, **quantum noise**, which affects the accuracy of calculations, and **scalability**, which is the difficulty of creating larger, more complex quantum systems capable of solving real-world problems.

However, to make this less pessimistic, both Google (2019) and Chinese scientists (2020) have announced quantum supremacy by performing a calculation that they claim is impossible to perform on classical computers in a reasonable amount of time. More about this breakthrough is described in Section 2.3.

In relation to the goals that IBM has set for itself, it can be assumed that beyond 2033, quantum-centric supercomputers will include thousands of qubits capable of running one billion gates, unlocking the full power of quantum computing.

5.1.9. In what situations are quantum computers more effective than classical ones, and in what situations are they less effective?

Quantum computers, at least in theory, are more efficient than classical computers at solving problems involving searching large state spaces, such as factoring large numbers, global optimization, and simulating complex quantum systems. Quantum algorithms, such as Grover's algorithm, can search unstructured databases much faster than classical algorithms.

The limitations of quantum computers include difficulties associated with their physical realization, such as decoherence and quantum errors, and the lack of practical quantum algorithms for many problems that are well-solved by classical computers. In addition, quantum computers require complex and expensive hardware to operate at extreme conditions, such as very low temperatures, which constitutes another challenge in their wide application.

5.1.10. What are the predictions for the future cooperation and coexistence of quantum and classical computers?

Following scenario can help to understand in a graphic way what quantum computers are, how they work, what their future is and whether classical computers will continue to coexist with them or will they be replaced:

The scenario is like a video game, where the ability to explore is determined by mathematical skills. In the pre-computer era, tasks like geometry and astronomy were terribly slow, like “walking.” The advent of computers created an opportunity to process larger data sets, expanding the mathematical capabilities of mankind, like the transition to “horseback riding.” However, certain areas remained inaccessible. Later technological advances introduced roads and cars onto this metaphorical map, creating a possibility to reach previously inaccessible regions. However, in this scenario, quantum computers are not simply faster vehicles; they are boats specially designed to navigate different terrains. These advanced systems form an opportunity to explore previously unexplored domains, tackling problems that classical computers cannot solve [\[68\]](#).

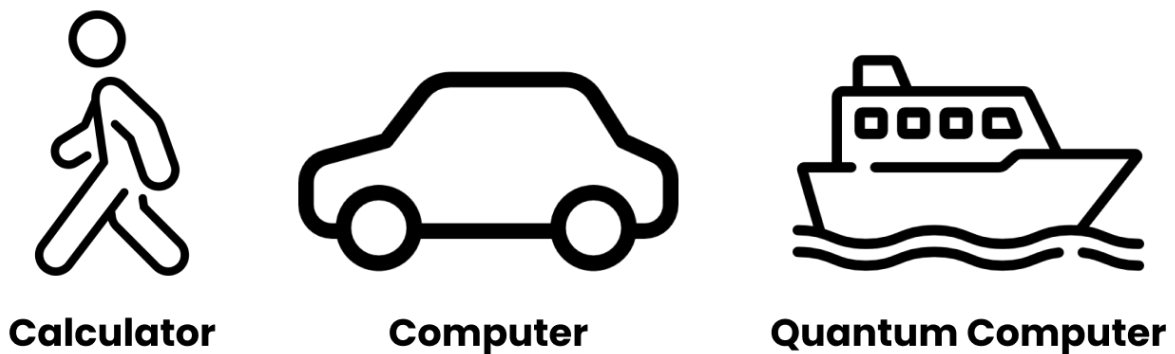


Figure 5.1: Coexistence of Quantum and Classical Computers (Created by the Author)

The above history clearly shows that quantum computers are not the successors of classical computers. They were not created to perform the same tasks faster, but for completely new, sometimes completely unknown tasks. They are still in the research and development phase, and their practical application is still limited. Although progress in this field is rapid, it can be stated with great confidence that they will coexist with the classical computers that are already known. It is also worth recalling the hybrid algorithms QAOA and VQE described in this work. They use a quantum computer to perform calculations, while the classical computer optimizes the parameters of the quantum algorithm.

5.2. IBM EAGLE 127-qubits processor.

At this stage it is also worth introducing the IBM Eagle processor [\[48\]](#), with the help of which the following issues were implemented. A quantum computer, with the power of one hundred twenty-seven qubits, which can be accessed via the IBM Quantum cloud platform.

The main assumption of this project was the practical implementation of the Shor and Grover algorithm and using them to evaluate the capabilities of modern quantum computers, present the results and compare them with the capabilities of classical computers.

When collecting information on the first of the mentioned algorithms, contradictory information may have been encountered. On the one hand: The largest integer that has been factored so far was twenty-one.

On the other hand: To factor an N -bit number, a quantum computer with stable computing power of $2N$ qubits are needed. So, to factor a five-bit number like twenty-one, a ten-qubit quantum computer is needed.

So why, in an era of easy access to hundred and twenty-seven qubits computers, are such unsatisfactory results still achieved?

Having access to a one hundred twenty-seven qubits IBM Quantum Eagle processor theoretically opens the possibility of implementing Shor's algorithm to factor numbers larger than five bits. However, there are significant limitations that could make it difficult or even impossible to do in practice.

5.2.1. Quantum Circuit Complexity

Shor's algorithm requires a complex quantum circuit that consists of numerous quantum gate operations and requires precise entanglement of many qubits. In the case of factoring a five-bit number, about ten or even fifteen logical qubits are needed, depending on the exact implementation. However, in practice, the number of physical qubits needed can be many times greater due to the need for error-correcting codes.

5.2.2. Qubit correlation and entanglement

Shor's algorithm requires efficient entanglement of qubits to perform large-scale quantum operations. In practice, the level of entanglement between qubits in quantum processors is limited and may not be high enough to ensure that the algorithm works correctly on many qubits. If the entanglement between qubits is not sufficiently stable and strong, the accuracy of the computations may be compromised.

5.2.3. Decoherence

Decoherence is one of the most serious problems in current quantum computers. Qubits can lose their quantum nature through interactions with their environment, leading to errors in calculations. The longer a calculation takes, the greater the risk of decoherence, which is a particular problem for complex algorithms such as Shor's algorithm.

5.2.4. Noise and errors in quantum operations

Modern quantum computers are burdened with noise, which can lead to errors in the operations performed. In the context of Shor's algorithm, even small errors can accumulate during the execution of calculations, which can result in incorrect results. The important level of noise and low fidelity of quantum gates are a serious limitation in the practical application of the algorithm.

5.2.5. Error correction

Quantum error correction is essential for scaling quantum computations to the level where Shor's algorithm can be practically implemented. Unfortunately, implementing efficient error correction requires many more physical qubits than logical qubits. A one hundred twenty-seven qubits processor may not be enough to efficiently implement Shor's algorithm with full error correction, which is an additional limitation.

5.2.6. Coherence time

Shor's algorithm requires that qubits maintain their coherence throughout the computation. In practice, the coherence time of available qubits is limited, which may prevent the algorithm from completing before decoherence significantly affects the result

5.2.7. Modern possibilities

IBM quantum computers, such as those with one hundred twenty-seven qubits, refer to the number of physical qubits that are available in the processor. However, the number of logical qubits, or qubits used for the actual calculations in quantum algorithms, is smaller due to the need for quantum error correction.

Quantum error correction requires many physical qubits (usually tens or hundreds) to represent one logical qubit. The exact number of physical qubits needed for one logical qubit depends on the specific error correction implementation, but for most current technologies, several dozen to several hundred physical qubits are required for one logical qubit.

Estimates of the number of physical qubits needed to realize one logical qubit vary, but a reasonable number is considered to be around one thousand. In their paper “How to Compute a 2048-bit RSA Integer in 8 Hours Using 20 million Noisy Qubits,” authors Craig Gidney and Martin Ecker estimate that it would take around twenty thousand logical qubits to break RSA encryption, which translates to around twenty million physical qubits. By their calculations, such a system could break RSA encryption in just eight hours. This is a task that even the most powerful supercomputers cannot achieve in any reasonable time, which clearly shows the potential of Shor’s algorithm.

However, the largest number that has been calculated using Shor’s algorithm is twenty-one, and the smallest is fifteen. This limited range is due to two main reasons:

First, Shor's algorithm requires a vast number of qubits, and such a number does not yet exist in practice.

Second, current qubits are very susceptible to noise, which leads to errors in the calculations. The high number of errors makes it impossible to distinguish correct results from incorrect ones [\[69\]](#).

Summary

Although in theory a one hundred twenty-seven qubits quantum processor, such as the IBM Eagle, could be used to factor numbers much larger than five-bits using Shor's algorithm, in practice there are many technological limitations. Problems related to decoherence, noise, qubit entanglement, and limited fidelity of quantum operations can make it significantly difficult or even impossible to correctly execute the algorithm on such many qubits. As a result, although factoring numbers of five-bits or less is possible under current conditions, computations on larger numbers are much more difficult to perform and require further advances in quantum technology.

6. Introduction to Shor's algorithm - History, field of usage, operating principles, use cases.

6.1. History

Shor's algorithm, developed by Peter Shor in 1994, is one of the most groundbreaking achievements in the field of quantum computing [\[11\]\[49\]](#). Its creation was a response to fundamental problems related to data security and the growing interest in the possibilities offered by quantum computing. The introduction of this algorithm revolutionized the approach to problems that were previously considered unsolvable using classical computational methods.

6.1.1. Scientific Context and Inspiring Problems

In the 1980s and 1990s, scientists began to realize that classical computers, while incredibly powerful, had their limitations, especially when it came to computational problems that required vast amounts of time and memory. One of the most famous such problems is factoring large numbers into prime factors, which underlies the security of many cryptographic systems such as RSA.

In the traditional approach to this problem, the time required to factor a number with many digits grows exponentially with its size. For exceptionally large numbers, this process becomes impractical, making factoring methods effective as a means of security. However, theoretical studies of quantum computers have pointed to the possibility of dramatically accelerating some computations, including factoring, thanks to phenomena such as superposition and quantum entanglement.

6.1.2. Creation of Shor's Algorithm

In 1994, Peter Shor, a mathematician and computer scientist working at Bell Laboratories, published a paper in which he presented a quantum algorithm that solves the integer factoring problem in polynomial time. This discovery gained immediate recognition in the scientific community because it showed that quantum computers could solve problems that were impossible for classical computers to solve in a reasonable amount of time.

Shor's algorithm is based on two key steps:

- first, using a quantum algorithm, it finds the period of the numerical function, which is a key element of factorization,
- and then uses classical calculations to find the prime factors of the number from that period.

By using quantum parallelism of computation, Shor's algorithm can reduce the factorization time from exponential to polynomial, which is a huge speedup compared to classical methods.

6.1.3. Significance and Implications of the Discovery

The creation of Shor's algorithm had far-reaching implications not only for the theory of quantum computing, but also for practice, especially in the field of cryptography. The fact that this algorithm could potentially break factorization-based security has raised considerable interest and concerns in the information security community.

The development of Shor's algorithm also contributed to accelerating research into the practical construction of quantum computers that could implement it. To this day, it remains one of the most famous and widely studied quantum algorithms, which not only demonstrated the potential of quantum computing, but also inspired further research into new algorithms and applications of quantum computers.

Summary

Shor's algorithm, discovered in 1994, is a milestone in the history of quantum computing. Its genesis is related to the search for efficient solutions to problems that are extremely difficult or impossible for classical computers to solve in practical time. The scientific context and problems related to cryptographic security were key factors that led to the creation of this algorithm. His discovery not only confirmed predictions about the computing power of quantum computers, but also opened new directions of research and challenges in the fields of quantum computing and cryptography.

6.2. Field of Usage

To fully understand the potential of Shor's Algorithm, it is necessary to explain the meaning of the **RSA** cryptographic algorithm.

RSA (Rivest–Shamir–Adleman) is one of the most used cryptographic algorithms, which is based on asymmetric cryptography. In asymmetric cryptography, two keys are used: public and private. The public key is used to encrypt a message, while the private key is used to decrypt it.

RSA is based on the mathematical problem of factoring large numbers. The main principle of RSA is that multiplying two large prime numbers is relatively easy, but it is difficult to recreate these numbers, having only their product. Thanks to this, the public key can be safely shared, because decrypting a message requires knowledge of the private key, which is associated with these prime numbers.

6.2.1. Cryptography

The most famous application of Shor's algorithm is in cryptography, more specifically in breaking encryption algorithms based on factoring large numbers. Classical cryptographic algorithms such as RSA exploit the fact that factoring large numbers is a computationally intractable problem for classical computers. Shor's algorithm allows factoring large numbers in polynomial time, which in turn makes algorithms such as RSA potentially vulnerable to being broken by a quantum computer.

6.2.2. The Future of Cryptography

The use of Shor's algorithm raises questions about the security of many current cryptographic systems. As quantum technology advances, there is a real risk that quantum computers will be able to break current encryption standards, which in turn will force the development of new cryptographic methods resistant to quantum attacks, such as post-quantum cryptography.

6.2.3. Applications in other fields

Although the main field of application of Shor's algorithm is cryptography, it has the potential to find applications in other fields where factoring large numbers plays a key role, such as mathematical and theoretical analysis, or in the optimization of certain types of problems. However, the most important and immediate practical effect of Shor's algorithm remains its ability to break factoring-based cryptosystems.

6.2.4. Qubit Number Requirements

Effective implementation of Shor's algorithm requires a quantum computer with enough stable qubits. For example, to factor a one thousand and twenty-four-bit number, theoretically about two thousand and forty-eight qubits would be needed. Current quantum computers, such as the IBM Eagle with one hundred twenty-seven qubits, can potentially manage factoring much smaller numbers, but implementing Shor's algorithm on a practical scale requires further technological development, particularly in the areas of qubit stability, error correction, and decoherence management. In summary, Shor's algorithm is fundamental to the future of cryptography and computer security. Its use in breaking classical encryption algorithms poses a serious challenge to data security in the post-quantum era.

6.3. Operating Principles

Its operation is based on two main elements: modular calculations and the quantum Fourier transform, which uses negative wave interference.

6.3.1. Modular calculations

The basis of Shor's algorithm is modular computation, which involves finding the remainder of a division to determine the period of some function. In the context of this algorithm, this computation is used to find the period r , which is crucial for factoring a number N . The algorithm randomly chooses a number a less than N and computes values $a^x \pmod{N}$ for various x . Essentially, this function has period r , such that $f(x) = f(x + r)$. Finding this period creates an opportunity to determine the greatest common divisor of N and $a^{r/2} \pm 1$, which leads to the factoring of N .

6.3.2. Quantum Fourier Transform and Negative Wave Interference

The second key element of Shor's algorithm is the Quantum Fourier Transform (QFT), which is a quantum version of the classical discrete Fourier transform, operating on quantum states. Shor developed an algorithm that uses QFT to transform a superposition of quantum states in such a way as to discover the periodicity of the function using a quantum measurement.

This transform operates on a superposition of quantum states, which allows for parallel processing of many possible outcomes at once, which is one of the fundamental advantages of quantum computing. It uses the phenomenon of negative wave interference, which allows for the suppression of inappropriate quantum states and the enhancement of those that correspond to the true period r . Thanks to this, after applying QFT, quantum states representing the correct period become more visible and can be extracted with high probability during measurement. In this way, QFT allows finding the periodicity in modular calculations, which is crucial for the next steps of the algorithm.

6.3.3. Iteration and measurement

Shor's algorithm works by initializing a quantum state in which a quantum register is prepared to hold all possible computational outcomes. Modular computations are then performed on these states, and the outcomes are transformed using a quantum Fourier transform.

The final step is to measure the quantum state, which, thanks to the amplification resulting from QFT and negative wave interference, returns a result that allows to determine the period r with high probability. From the found period, it is then possible to determine the prime factors of N .

Summary

Shor's algorithm, by using modular computation and quantum Fourier transform, enables fast factorization of large numbers. The use of negative wave interference in QFT is a key element that allows finding the period of the function, which leads to solving the factorization problem. Thanks to this, Shor's algorithm poses a serious threat to classical cryptographic systems based on the difficulty of factoring numbers.

7. In-depth analysis of Shor's algorithm - Theoretical possibilities of chosen algorithm.

The mathematical, physical and quantum Fourier transform fundamentals, important for the theoretical foundations of Shor's algorithm, have already been covered in previous topics [\[11\]\[49\]](#). However, there are still many issues worth analyzing:

7.1. Computational complexity

One of the most important aspects of Shor's algorithm is its computational complexity. Classical factoring algorithms, such as Pollard's rho algorithm or the quadratic sieve (QS) algorithm, have exponential complexity with respect to the number of bits n representing the number N . This means that the time taken to factor the number grows very quickly with the number of bits. In contrast, Shor's algorithm runs in polynomial time, meaning the time taken is polynomial in $\log N$, where N is the size of the integer given as input. Specifically, it takes quantum gates of $O((\log N)^2(\log \log N)(\log \log \log N))$ using fast multiplication or even $O((\log N)^2(\log \log N))$ utilizing the asymptotically fastest multiplication algorithm currently known due to Harvey and Van Der Hoven. In practice, implementing the algorithm requires solving many engineering problems. Until now, the largest factorization using Shor's algorithm was a number consisting of only a few bits, which is due to current technological limitations, such as qubit decoherence and imperfections in the implementation of quantum operations.

7.2. Quantum Circuit Optimization

One of the key elements of the successful implementation of Shor's algorithm is the optimization of the quantum circuit, which allows minimizing the number of required quantum operations (gate depth) and reducing the impact of quantum errors. There are studies on ways to optimize such circuits, especially in the context of reducing the number of controlled gates, which are expensive in terms of quantum resources.

The most important ways to optimize a quantum circuit:

7.2.1. Reducing the number of quantum gates

Reducing the number of quantum gates is one of the most direct approaches to optimization. Each additional gate increases the risk of introducing errors, so the key goal is to minimize the number of operations required to complete the task.

- **Simplifying Controlled Gates:** **Controlled gates** (for example Toffoli gate or CNOT) are particularly resource intensive. There are optimization techniques that convert more complicated controlled gates into combinations of simpler operations, which allows reducing the total number of operations in the circuit.
- **Compiling quantum circuits:** Advanced quantum compilers can transform an original circuit into an optimized version of it by identifying redundant operations, simplifying gate sequences, and minimizing the number of qubits needed for the implementation.

Controlled gates

An essential element of the circuit is also controlled gates, such as CNOT gates (controlled NOT) and controlled phase gates. These gates allow for the entanglement of quantum states between different qubits, which is necessary for the correct execution of quantum operations in Shor's algorithm. Controlled gates allow for the dependence of one gate on the state of another qubit, which allows for the implementation of complex logical operations.

7.2.2. Qubit Locality and Routing

In practical implementations of quantum circuits, qubits can be physically constrained to interact only with nearby qubits. Optimization involves arranging qubits and gates (adapting the topology of the quantum circuit) to minimize the number of information transport operations (routing) between distant qubits.

7.2.3. Time optimization (gate depth)

Gate depth is the number of steps (gate layers) needed to complete a circuit and is crucial to the circuit's performance. Reducing gate depth reduces the time it takes for qubits to maintain coherence, which is critical for computational accuracy. This is accomplished by performing operations on qubits that are independent of each other in parallel.

7.2.4. Quantum error correction, scalability

Scalability of Shor's algorithm in the context of factoring larger numbers requires the availability of stable qubits with high fidelity and long decoherence. Current research on the scalability of the algorithm focuses on reducing the number of physical qubits needed to implement a logical qubit and on error correction methods that can significantly increase the number of physical qubits needed to implement the algorithm.

Quantum error correction is essential for the efficient implementation of large quantum circuits. Optimization in this area involves minimizing the number of physical qubits needed to realize one logical qubit and optimizing **error-correcting codes**.

Error-correcting codes: Selecting the appropriate error correcting code from among the following:

- stabilizer codes:
 - surface code,
 - CSS.
- topological codes (torus code)
- Bacon-Shor code - hybrid of above ones (CSS and torus)
- Low-density quantum codes
 - Hyperbolic Code

allow for increased fault tolerance of the circuit at minimal cost in the number of qubits.

7.2.5. Quantum data compression

Quantum data compression is a technique that reduces the number of qubits needed to represent a quantum state without losing essential information.

- Appropriate information encoding: Using advanced quantum coding techniques can reduce the number of qubits needed to process complex data, leading to fewer gate operations and increased circuit efficiency.

7.3. Comparison with classical methods

In terms of theory, Shor's algorithm significantly outperforms classical factoring methods, especially for large numbers. Classical algorithms, such as the aforementioned Pollard's rho or QS, have exponential complexity and in practice become inefficient for exceptionally large numbers, which are commonly used in cryptography. For this reason, Shor's algorithm has the potential to break security based on such methods, which poses a challenge to modern cryptographic systems.

7.3.1. Application of Hadamard Gates

A particularly crucial step is also to introduce the quantum register into a superposition state of all possible input values. This is achieved by applying Hadamard gates to all qubits in one of the registers. Each Hadamard gate transforms the state $|0\rangle$ of the qubit into a uniform superposition of states $|0\rangle$ and $|1\rangle$, which causes the entire register to take on a superposition of all possible binary states. It is a key element of many quantum algorithms, enabling parallel processing of information in quantum space.

The Hadamard Gate (denoted H) is a fundamental quantum operation that transforms a single qubit state into a superposition state. Formally, the Hadamard gate transforms the state $|0\rangle$ into $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$, and the state $|1\rangle$ into $\frac{|0\rangle-|1\rangle}{\sqrt{2}}$. This allows a qubit that is initially in one of the specified states ($|0\rangle$ or $|1\rangle$) to enter a simultaneous superposition of both states with equal probability after applying the Hadamard gate. The Hadamard gate is described by the matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

It is a key element of many quantum algorithms, enabling parallel processing of information in quantum space.

8. Implementation and use of Shor's algorithm - Practical possibilities, performance.

When the topic of authoring the paper was created, the goals and objectives of the tests were to demonstrate the highest possible number that could be factorized using Shor's quantum algorithm. To do this, it was necessary to implement a practical implementation of this algorithm and to conduct time-based tests. The following section is based on the following bibliography: [\[11\]\[49\]\[50\]\[51\]\[52\]\[53\]](#). The code was developed with the help of PSNC's quantum computing staff.

8.1. Practical Implementation

Step one: Selecting the number to factorize 'N'.

Shor's algorithm is used to find the factors of an integer N . N must be the product of two prime numbers (for a number composed of more than two prime numbers, it is less efficient).

Step two: Select the number 'a'.

Choose a random number 'a', one that satisfies the inequality $2 < a < N$ and $\gcd(a, N) = 1$. If $\gcd(a, N) \neq 1$, then $\gcd(a, N)$ is one of the factors of N , and there is no need to continue with the algorithm.

Step three: Create a quantum circuit to make a modular exponentiation.

Modular exponentiation is a key component of Shor's algorithm. It requires designing a quantum circuit that calculates $a^x \pmod{N}$ for different values of x .

Step four: Implementation of the quantum periodic algorithm.

Shor's algorithm uses the Quantum Fourier Transform (QFT) to find the period of a function $f(x) = a^x \pmod{N}$.

Steps:

1. Create a quantum register: use qubits to represent possible x values.
2. Apply Hadamard gates: Enter the superposition of all possible values of x .
3. Modular exponentiation: calculate the function $a^x \pmod{N}$ on qubits.
4. QFT: Perform the inverse Fourier transform (QFT) on qubits.
5. Measurement: Measure the quantum state to get the result.

Step five: Analysis of measurement results

After Shor's algorithm is executed, the results of the measurements are analyzed to find the period of the function $f(x)$. The period found is key to decomposing N into factors.

Simulator: The measurement results are usually in the form of probability distributions that can be easily interpreted.

Actual quantum computer: Results may be subject to noise, which may require additional statistical analysis.

Step six: The classic part of the algorithm

After finding the period, the algorithm goes to the classical part, which calculates the greatest common divisor of $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$. If these values are different from one and N , they are factors of the number N .

Step seven: Run the circuit on a quantum simulator.

For testing and debugging the algorithm, it is an innovative idea to first run it on a quantum simulator. This is useful because simulators allow fast circuit execution without the constraints of physical hardware and queued task waiting.

Step eight: Run the circuit on a real quantum computer.

To run the algorithm on a real quantum computer, it is necessary to select a suitable backend, configure it and run the circuit.

Examples of backends: *ibm_brisbane*, *ibm_kyoto*, *ibm_kyiv*, *ibm_sherbrooke*.

Step nine: Process results and determine factors.

The ultimate goal of the algorithm is to determine the factors of the number N . Based on the measurement results, these factors can be calculated and checked for correctness.

Simulator: Analysis of the results is simpler and less subject to error.

Real quantum computer: The results may require additional processing to reduce the impact of noise.

8.1.1. Implementation

Classical approach's implementation

```
1.  select N
2.  a = rand(2, N)
3.  if gcd(N, a) > 1: finish
4.  v = [], q = 1
5.  while q != 1:
6.    q = q*a mod N
7.    v.append(q)
8.    if len(v) mod 2 != 0: goto 2
9.    p = v[len(v)/2]
10.  if p+1 == N: goto 2
11.  return gcd(p+1, N), gcd(p-1, N)
```

Quantum approach implementation

```
1.  qc = QuantumCircuit
2.  apply Hadamard gates
3.  apply X gates - (Qubit flip)
4.  apply modular exponentiation
5.  apply QFT.inverse()
6.  connect to simulator/backend
7.  compile & run circuit
8.  get result.counts()
9.  perform classical post-processing to find prime factors
```

8.2. Performance Analysis

The proof on Shor's algorithm clearly shows the poly-logarithmic acceleration of number factorization with the quantum approach relative to the exponential complexity of the classical approach. The main aspect enabling such acceleration is the representation of N numbers in the binary system, which becomes extremely efficient when using qubits. In addition, a key aspect is the use of QFT, whose complexity is $O((\log N)^2)$. When implementing the algorithm, it is also important to remember to use the appropriate number of qubits. It affects two factors - the speed of computation and fault tolerance. As the number of qubits increases, the speed of finding the space of potential solutions increases, but so does the susceptibility to computational errors and noise.

8.2.1. Expectations

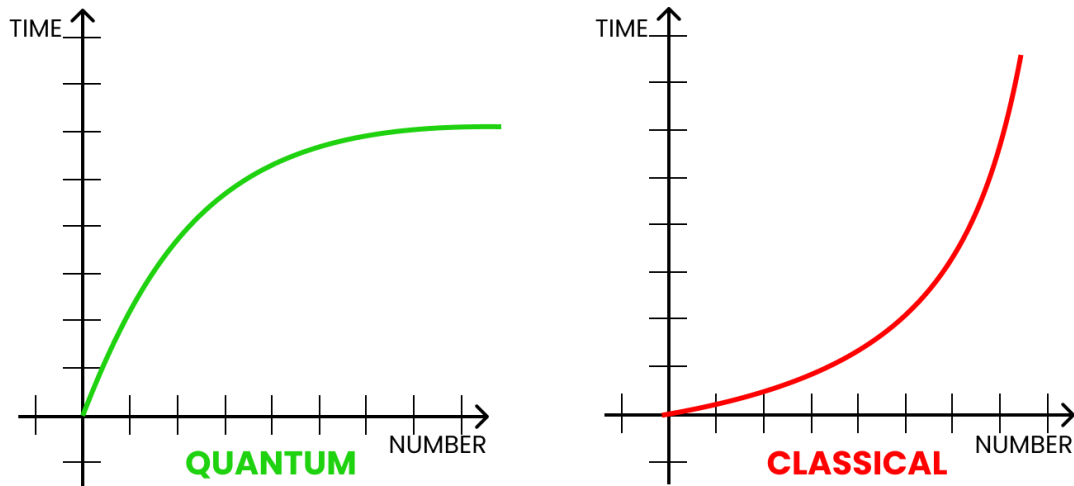


Figure 8.1: Performance Expectations (Created by the Author)

8.2.2. Reality

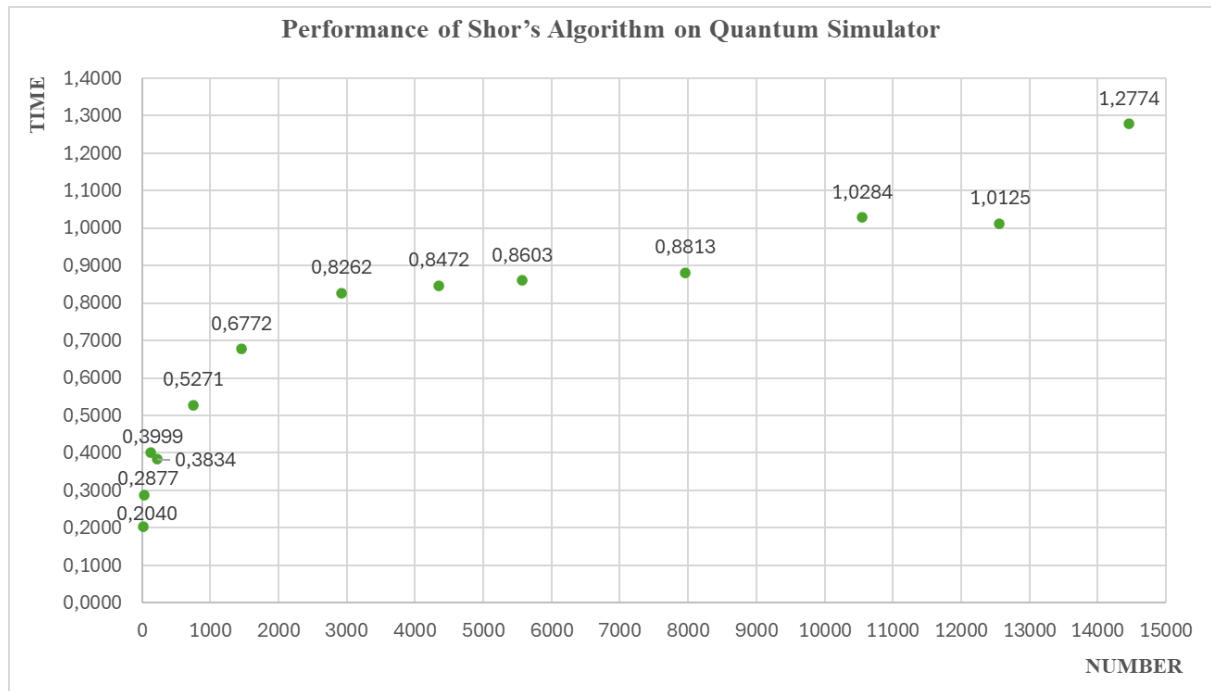


Figure 8.2: Performance of Shor's Algorithm on Quantum Simulator (Created by the Author)

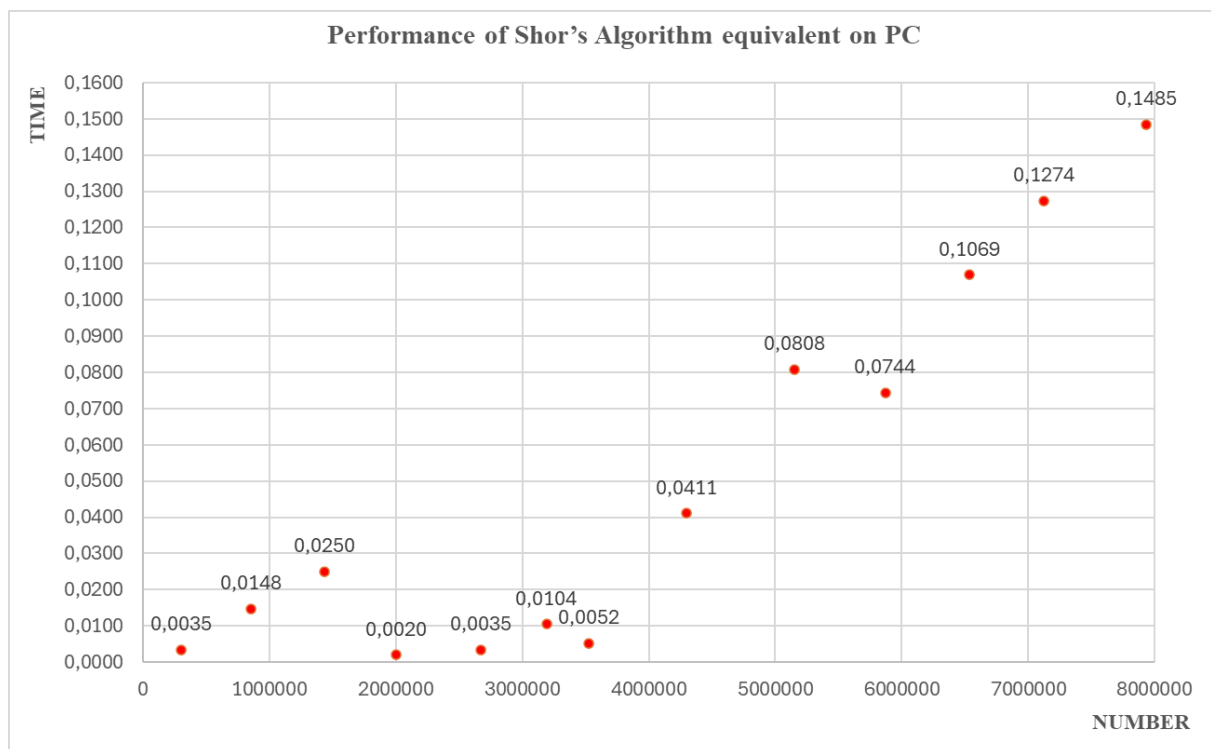


Figure 8.3: Performance of Shor's Algorithm equivalent on PC (Created by the Author)

8.3. Real-world Applications

The implementation of the Shor algorithm is potentially groundbreaking for modern cryptography, especially in the context of breaking encryption systems based on factorization of large numbers, such as RSA. In theory, Shor's algorithm could factorize large numbers, which currently underpins the security of many cryptosystems, making them vulnerable to attacks using quantum computers. Its successful implementation on scalable quantum computers could force a shift to new quantum-secure encryption methods to protect data. However, as of today, there is no infrastructure capable of providing such computing power and noise immunity.

8.4. Discussion about the results

Performance and timing tests were conducted on all three available environments, a classical computer, a quantum simulator on a classical computer, and a quantum QPU, thanks to IBM's provision of remote access to the platform. The results of the first two are presented in the section above (Figure 8.2 and Figure 8.3), while the results from quantum computing will not be shown in the chart. Despite trials specifically conducted during low-traffic hours on the platform, a significant part of the measured times is the submission of a request to the backend and the queuing of a task, which has an unreliable effect on the measured results. The averaged measurements are shown below (Table 8.1). Neither the size nor the order of the number has any effect on the length of the measurement. In contrast, the graphs above show the trends described in the expectations (Figure 8.1). Classical calculations, with minor exceptions - resulting from small measurements that are prone to retain exponential complexity.

Number	Avg time [s]
15	33,4075
21	42,3030
33	36,9073
35	52,9890
51	45,2059
133	47,1970
177	44,0420

Table 8.1: Results from a quantum computer (Created by the Author)

Below is a snapshot from the IBM platform showing approximate execution times for the assigned tasks.

✔ Completed	26 Aug 2024	26 Aug 2024	13s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	13s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	13s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	13s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	14s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	7s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	6s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	6s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	6s	Session	ibm_kyoto
✔ Completed	26 Aug 2024	26 Aug 2024	6s	Session	ibm_kyoto

Figure 8.4: Snapshot from the IBM platform (Created by Author)

The results do not equal expectations, since the intention of the thesis was to factorize much larger numbers. On the other hand, these are results that show the dynamic development of the world of quantum computers, because so far, the highest number officially subjected to successful factorization using a quantum computer was the number twenty-one. In this thesis, the number one hundred seventy-seven was subjected to fruitful decomposition. However, it is worth mentioning that out of ten attempts made, only four were successful. Such results give cause for optimism, while at the same time leaving considerable ground for the development of the quality and efficiency of quantum computers.

9. Introduction to Grover's algorithm - History, field of usage, operating principles, use cases.

9.1. Brief history:

Grover's algorithm was proposed in 1996 by Indian-American physicist and computer scientist, Lov Grover, working at Bell Labs. It is one of the fundamental algorithms in the field of quantum computing and, along with Shor's algorithm, is considered one of the most important achievements in the field [\[12\]\[54\]](#).

9.1.1. Research context

In the 1990s, the quantum computer was still a theoretical concept, but more and more scientists were beginning to see its potential. At that time, traditional classical computers (built on the principles of classical mechanics) were already widely developed, but were approaching some fundamental limitations, such as in terms of data processing speed and the ability to solve certain tasks in a reasonable amount of time.

The first theoretical foundations of quantum computing were laid in the 1980s by people such as Richard Feynman and David Deutsch to whom Chapter 2.3 of this paper is dedicated. One of the breakthroughs was the discovery in 1994 by Peter Shor of an algorithm that could theoretically decompose numbers into prime factors in polynomial time, which would be significant for cryptography.

In this research context, Grover presented his algorithm, which solved the problem of searching an unstructured database in quadratic time relative to the number of elements, a significant improvement over the classical algorithm, which would require linear time. This meant that for a database with N elements, Grover's algorithm would require about $\sqrt{N}$ operations, instead of N operations as in the case of classical computers.

9.1.2. The discovery and importance of the algorithm

Grover's algorithm has shown that quantum computing could speed up problem solving not only in cryptography, as Shor's algorithm did, but also in a wide range of applications such as database searching, optimization and artificial intelligence. Grover's algorithm proved that even for unstructured problems for which there is no clear structure that can facilitate searching, quantum computing can offer significant speed-ups compared to classical methods.

The discovery was important for the further development of quantum computing, demonstrating the potential of quantum computers in solving a wide range of problems. Grover's algorithm remains to this day one of the most important examples of the use of the principle of superposition and quantum interference in quantum algorithms.

9.2. Field of Usage:

9.2.1. Searching unstructured databases

- **Searching in large databases:** In cases where databases are characterized by a vast number of records and a lack of internal structure, Grover's algorithm makes it possible to significantly speed up the process of finding information, such as in medical, financial or customer data databases.
- **Searching encrypted data:** The algorithm can be used to quickly search encrypted databases, which is particularly important for maintaining security where traditional search methods prove too slow.

9.2.2. Optimization

Grover's algorithm is also used in solving optimization problems in which the goal is to find the best solution in a large set of possible solutions. In such cases, the algorithm can significantly speed up the process of finding the optimal solution. As can easily be seen, its primary task is still searching, but the context of application changes. Examples of applications include:

- **Logistics optimization:** An algorithm can be used to find the most efficient transport route from among many possible variants.
- **Optimization in engineering:** Can be used to find the best structural or material design that meets specific criteria.

9.2.3. Artificial intelligence & machine learning

Grover's algorithm can be used to speed up some of the processes involved in searching the solution space in artificial intelligence and machine learning. This can include searching for optimal hyperparameters, feature selection or solving classification problems. Examples of applications include:

- **Feature selection:** Grover's algorithm can be used to quickly select the most relevant features from a large dataset, which is crucial in machine learning.
- **Model Optimization:** The algorithm can speed up the process of optimizing the hyperparameters of artificial intelligence models, which is important for the effectiveness of these models.

9.2.4. Cryptography

Grover's algorithm can be important in the context of cryptography. On the one hand, it allows speeding up the process of cracking passwords or encryption keys by searching the entire space of possible keys, which is a potential threat to classical cryptographic algorithms. On the other hand, it can be used to evaluate the security of various cryptographic schemes.

Examples of applications include:

- **Security testing of cryptographic algorithms:** Grover's algorithm can be used to assess the cryptographic strength of keys and identify potential vulnerabilities.

9.2.5. Solving decision problems

Grover's algorithm can also be used to quickly solve decision-making problems in which it is necessary to search a large space of solutions to answer a “yes/no” question.

Examples of applications include:

- **Checking the correctness of solutions:** Grover's algorithm can speed up the process of verifying the correctness of solutions in problems like SAT (Satisfiability of Logical Formulas).
- **Finding optimal solution:** The algorithm significantly speeds up finding the best solution in a large set, such as in logistics, helping optimize routes in terms of cost and time.
- **Pattern recognition and classification:** Grover's algorithm is used for faster pattern recognition in data, such as identifying key DNA sequences in genomics, which boosts decision-making processes in medical diagnostics.

9.3. Operating Principles

9.3.1. Principle of amplitude amplification

Amplitude amplification is a key technique used in Grover's algorithm to focus the quantum superposition of states toward the state representing the sought-after element. The operation of the algorithm can be described in several steps:

1. **Initial superposition:** The algorithm begins by initializing the quantum register in a state of superposition of all possible input states. This is achieved by applying a **Hadamard gate** to each qubit, causing each qubit to assume a superposition state in which all possible bit configurations have equal amplitude.
2. **Oracle operation:** Next, the so-called **Oracle operation** is applied, which is a **unitary quantum operation** that modifies the amplitude of the state corresponding to the searched element. Oracle

recognizes the searched element and changes its phase (multiplies the amplitude of the state by minus one), while the amplitudes of the other states remain unchanged.

3. **Diffusion operation** (inversion around the mean): The next step is to apply a diffusion operation, also known as inversion around the mean, which increases the amplitude of the state corresponding to the sought-after element at the expense of reducing the amplitudes of the other states. This operation works by reflecting the amplitudes of all states around the average amplitude of all states. The phenomenon of interference is used to constructively amplify the amplitude of the wanted state (thanks to the diffusion operation) and destructively extinguish the amplitudes of the other states.
4. **Iteration:** steps two and three are repeated, with each iteration increasing the probability that a measurement of the quantum register will return a state corresponding to the desired element. The number of iterations required is of the order of $\sqrt{N}$, where N is the number of possible states (that is, the size of the state space).

It is worth recalling at this stage that **Hadamard gates** are a key element of many quantum algorithms, enabling parallel processing of information in quantum space. Shor's algorithm also makes use of this step, and it is on this occasion that Hadamard gates are described and explained in more detail.

9.3.2. State space search efficiency.

Grover's algorithm is particularly efficient in searching the state space compared to classical algorithms. In the case of classical search methods, in the worst-case scenario it is necessary to go through every element in the database, which takes linear time, proportional to the number of N elements.

Grover's algorithm, on the other hand, reduces this time to $\sqrt{N}$, which means that the number of operations needed to find a searched element is proportional to the square root of the number of elements. This increase in efficiency is due to the use of the principles of quantum mechanics, in particular superposition and interference, which allow multiple possible solutions to be processed simultaneously.

Other key elements that affect the performance and efficiency of Grover's algorithm:

9.3.3. Unity of operations

Each operation in Grover's algorithm, including **Oracle** and **diffusion**, is represented by unitary quantum operators that preserve the quantum state norm. Singularity means that these operations are reversible and do not cause loss of information, which is crucial for maintaining quantum consistency throughout the computational process. As a result, Grover's algorithm can efficiently manipulate quantum states and control amplitudes, leading to a successive increase in the probability of finding the correct result.

9.3.4. Oracle operation as a black box

The Oracle in Grover's algorithm is treated as a so-called black box, which means that its internal structure is not important for the algorithm's operation itself. All that is important is that the Oracle can correctly recognize the state being sought and change its phase accordingly. This structure makes it possible to apply Grover's algorithm widely to various problems, provided that a suitable Oracle function can be constructed.

9.3.5. Iteration Limitation Principle

The efficiency of Grover's algorithm depends on the appropriate choice of the number of iterations (known as Grover's steps). Too few iterations may not be enough to achieve sufficient amplification of the amplitude of the sought state, while too many iterations may cause “over-optimization,” meaning the weakening of the amplitude of the sought state due to further interference. The optimal number of iterations is about $\frac{\pi}{4}\sqrt{N}$, where N is the number of possible states. This rule of thumb is important for the algorithm to operate with maximum efficiency.

The fundamentals of Grover's algorithm include not only the principle of amplitude amplification, but also quantum superposition and interference (concepts that have already been cited many times in this paper and do not require additional translation), the singularity of operations, the principle of limiting the number of iterations, and the concept of the Oracle as a black box. All these elements work together to create an algorithm that is not only efficient in searching the state space, but also versatile and flexible in its application to various optimization problems.

10. In-depth analysis of Grover's algorithm - Theoretical possibilities of chosen algorithm.

This chapter provides an in-depth analysis of Grover's algorithm, covering quantum circuit design, initialization of quantum states, the oracle operator, the diffusion operator, iteration of the Grover operator, and measurement [\[12\]\[54\]](#).

10.1. Quantum Circuit Design

Describing the quantum circuit used to implement Grover's algorithm, discussing key operations (for instance: Grover's operator, reflections).

The quantum circuit for implementing the Grover algorithm consists of several key components that work together to search the state space and increase the probability of finding the desired element. The following is a description of this circuit and a discussion of the key operations.

10.1.1. Quantum state initialization

Grover's algorithm begins with the preparation of a quantum register consisting of n qubits, where $N = 2^n$ is the number of possible states in the search space. At first, each qubit is initialized in the $|0\rangle$ state. Then a Hadamard gate is applied to each qubit to put it into a superposition state:

$$H^{\otimes n}|0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

After this operation, the quantum state of the register represents a superposition of all possible input states.

10.1.2. Oracle operator

The Oracle (or phase operator) is a key component of Grover's algorithm, responsible for identifying the desired state. Oracle is a unitary quantum operation O_f , which operates on a state $|x\rangle$ as follows:

$$O_f|x\rangle = \begin{cases} -|x\rangle & \text{jeśli } x = x_s \\ |x\rangle & \text{w przeciwnym razie} \end{cases}$$

Oracle changes the phase of the x_s state corresponding to the item being searched for, leaving the phases of other states unchanged. In practice, Oracle can be implemented using a variety of techniques depending on the specific problem being solved.

10.1.3. Diffusion operator (inversion around the mean)

When Oracle is applied, the state of the sought-after element has an altered phase. To increase its amplitude, a diffusion operator, also called inversion around the mean, is applied. This operator can be represented as:

$$D = 2|\psi\rangle\langle\psi| - I$$

Where $|\psi\rangle$ is the superposition state after applying Hadamard gates, and I is the identity operator. It is worth noting that the diffusion operation (inversion around the mean) is a reflection around the initial state of the superposition. The initial state, reached after applying Hadamard gates, is a kind of “axis” for this operation. The diffusion operation acts on all states, bouncing the amplitudes of the quantum states around the average amplitude, resulting in an amplification of the amplitude of the desired state and a weakening of the amplitudes of the other states. In practice, each iteration of the Grover operator shifts the amplitudes closer to values that maximize the probability of finding the wanted state when measured.

Interpretation of reflection geometry

The operation of Grover's algorithm can also be understood from the perspective of vector geometry in Hilbert space, in which quantum operations can be interpreted as rotations of vectors. Oracle changes the phase of the vector corresponding to the desired state, which in geometric terms can be thought of as a reflection of the vector relative to a hypersurface perpendicular to the vector representing the desired state. The diffusion operator then bounces the resulting vector around the initial superposition state. These subsequent reflections move the state vector closer to the vector corresponding to the desired state, increasing its amplitude.

Balance between constructive and destructive interference

Grover's algorithm uses the principle of quantum interference, where amplitudes can add up (constructive interference) or cancel out (destructive interference). Oracle causes the amplitude of the wanted state to change sign, leading to constructive interference in subsequent iterations of the diffusion. This increases the probability of measuring the wanted state, while the amplitudes of the other states are gradually reduced.

10.1.4. Iteration of the Grover operator

The Grover operator, which combines the Oracle operator and the diffusion operator, is used repeatedly to increase the probability of detecting the desired state. The number of iterations of the Grover operator is a key element affecting the efficiency of the algorithm. With too few iterations, the amplitude of the sought-after state may not reach its maximum value, reducing the probability of a correct measurement result. On the other hand, too many iterations will cause the amplitude of the wanted state to start decreasing, leading to a situation where the probability of measuring the correct result decreases again. Therefore, it is important to precisely select the number of iterations for a particular problem. The number of iterations depends on the number of N states and is about $\frac{\pi}{4}\sqrt{N}$.

Each iteration of the Grover operator includes the following steps:

1. Application of the Oracle to a quantum state to change the phase of the sought state.
2. Application of the diffusion operator to amplify the amplitude of the sought state.

10.1.5. Measurement

After a sufficient number of iterations, the quantum register is in a state where the probability of reading the desired state is maximum. The final measurement of the quantum register returns a state with $|x_s\rangle$ high probability, which corresponds to solving the problem.

Summary

The quantum circuit of Grover's algorithm consists of the initialization of the quantum state, the application of the Oracle operator, the diffusion operator, and the repeated application of the Grover operator. Each of these components plays a key role in searching the state space and increasing the probability of finding the desired element. The Grover algorithm, due to its structure, can efficiently solve quadratically complex search problems faster than classical search algorithms.

10.2. Challenges and Limitations

The Grover algorithm, which is one of the most important quantum algorithms, offers a remarkable acceleration compared to classical methods for searching unstructured databases. However, its practical implementation faces numerous challenges and limitations, which are mainly caused by the number of available qubits, the precision of the calculations and the difficulty of scaling.

10.2.1. Universality and limitations of the algorithm

Although Grover's algorithm is extremely effective for problems that require searching unstructured spaces, its application is limited to cases where the state space is unstructured and difficult to access with classical methods. The algorithm does not speed up searches in the case of data structures that can be effectively searched by classical algorithms, such as searching in binary trees or using hashing.

10.2.2. The Impact of the Number of Qubits on the Speed of the Algorithm

In theory, the number of iterations required to achieve a high probability of finding the desired element is $\frac{\pi}{4}\sqrt{N}$. This means that as the number of qubits increases, the number of iterations also increases, which affects the execution time of the algorithm. However, increasing the number of qubits is associated with a greater risk of quantum errors and decoherence problems, which can negatively affect the accuracy of the calculations.

10.2.3. Number of qubits, scalability, and optimal number of qubits

The basic requirement of Grover's algorithm is an adequate number of qubits to represent all possible input states. For a database with N elements, the number of qubits n must be $n = \log_2(N)$.

For example, to search a database of one million elements, twenty qubits ($n = \log_2(10^6) \approx 20$) are needed. This number increases with the size of the database, which on the one hand increases the computational power of the algorithm, but on the other - the complications of its physical implementation.

The optimal number of qubits depends on the specific problem and the available quantum technology. Although an increase in the number of qubits theoretically increases the capabilities of the algorithm, in practice one encounters limitations related to control and stability. For example, to search a database with a billion elements, about thirty qubits are required, and the number of iterations is about 25,000 ($\frac{\pi}{4}\sqrt{10^9} \approx 25000$). Such parameters place high demands on the stability and precision of a quantum computing system.

10.2.4. Calculation precision and quantum errors

The precision of quantum operations is crucial to the successful operation of the Grover algorithm. The algorithm is extremely sensitive to errors in the execution of quantum operations, such as quantum gates and phase operations. Even small deviations can lead to erroneous results. This problem becomes more acute as the number of iterations and qubits involved increases, as the cumulative impact of errors can effectively cancel out the benefits of the Grover algorithm.

10.2.5. Technological limitations

Currently available quantum computers have a limited number of stable qubits, which limits the ability to implement Grover's algorithm to relatively small-scale problems. In addition, the precision of quantum gates is limited, which means that each operation introduces a certain level of error. As the number of qubits and iterations increases, the cumulative impact of these errors can significantly reduce the efficiency of the algorithm.

Summary

Despite its promising capabilities, Grover's algorithm faces serious challenges related to the number of qubits required, computational precision and scalability. Although in theory the number of qubits and iterations increases the search capability, in practice this leads to problems with quantum errors and decoherence. Therefore, to fully realize the potential of Grover's algorithm, it is necessary to further improve quantum technology, especially in the stability and control of qubits and the precision of the operations performed.

11. Implementation and use of Grover's algorithm - Practical possibilities, performance.

Grover's algorithm [\[53\]\[54\]\[55\]\[56\]\[57\]](#), one of the fundamental quantum algorithms, has revolutionized the approach to solving search and optimization problems, offering quadratic speedups compared to classical methods. This chapter discusses the implementation of Grover's algorithm and its applications in cryptography, optimization, machine learning, and other fields. This algorithm illustrates the power of quantum computing, pointing to the need for further development of technologies resistant to quantum attacks and exploration of new optimization possibilities.

11.1. Practical Implementation

Step one: Prepare the environment.

The first step is to prepare the environment by installing the appropriate set of *Qiskit* libraries. They provide the tools needed to create, simulate, and run quantum algorithms on both simulators and real quantum hardware.

List of necessary package libraries *qiskit*: *qiskit*, *qiskit-aer*, *qiskit-ibm-runtime*.

Step two: Configuring Grover's Algorithm and Building the Circuit

The first step of the implementation is to create an Oracle, which is a quantum circuit that is used to identify a specific state corresponding to the searched keys. The Oracle reverses the sign of the amplitude for the desired states, making them recognizable by the algorithm. Then, the *diffusion operator* and *phase query gates* are used, which complete the construction of the circuit for Grover's algorithm.

Step three: Running the circuit on the simulator.

It is worth running the circuit simulation on a specially prepared simulator first. This gives the opportunity to run the program locally, which provides the chance to test the circuit and check its correctness without using up the computational and time resources of the backend. If the result is correct, it is time for the next step to change the launch parameters.

Step four: Running the Circuit on a Real Quantum Computer

After authorization to the IBM Quantum platform, the user gains access to their computing resources. The connection must be properly configured, and the circuit must be carefully prepared for a specific QPU processor. Each available from the list has slightly different parameters than the others, which is why it is so important to prepare calculations for a specific model to obtain the best possible performance.

Step five: Analyzing the results.

The final step is to analyze the results obtained from simulations or actual quantum calculations. The results can be read from a 'dictionary,' where the key is the potential outcome, and the value is the probability of the outcome.

Visualization

The data presented in the form of a histogram shows the probability distribution of the searched area. If Grover's algorithm has been properly implemented, the most frequently occurring result will be the searched key.

11.2. Performance Analysis

The analysis of Grover's algorithm's performance clearly indicates a slower increase in the time needed to perform the same calculations as in the classical approach. While in classical calculations the complexity is $O(N)$, in quantum calculations it drops to $O(\sqrt{N})$.

It is also worth explaining where the difference between the declared linear complexity and the results leading to exponential complexity comes from. In these studies, it refers to n - meaning the number of bits needed to write the number N , so it can be described as $N = 2^n$. Hence the expected complexity of $O(2^n)$ and $O(\sqrt{2^n})$ for classical and quantum processing, respectively.

All trials for each value were performed twenty times and the results were averaged to better reflect the actual results.

11.2.1. Expectations

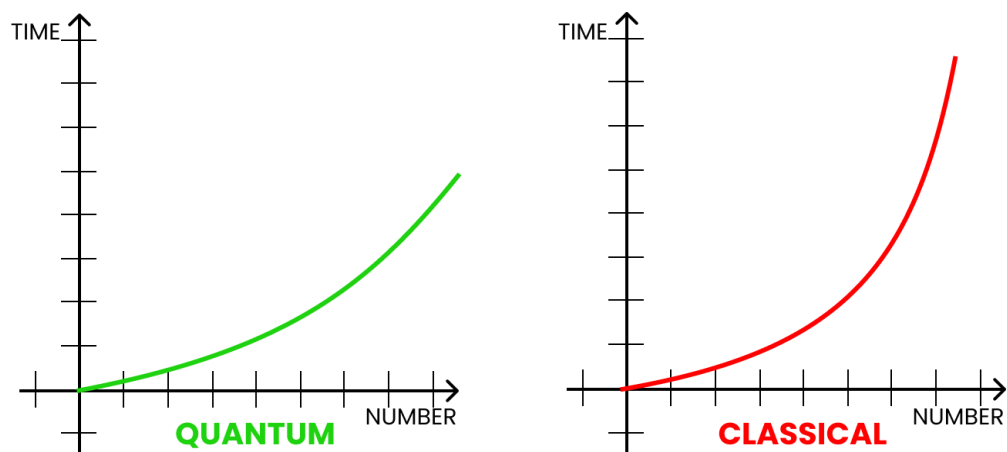


Figure 10.1: Performance Expectations (Created by the Author)

11.2.2. Reality

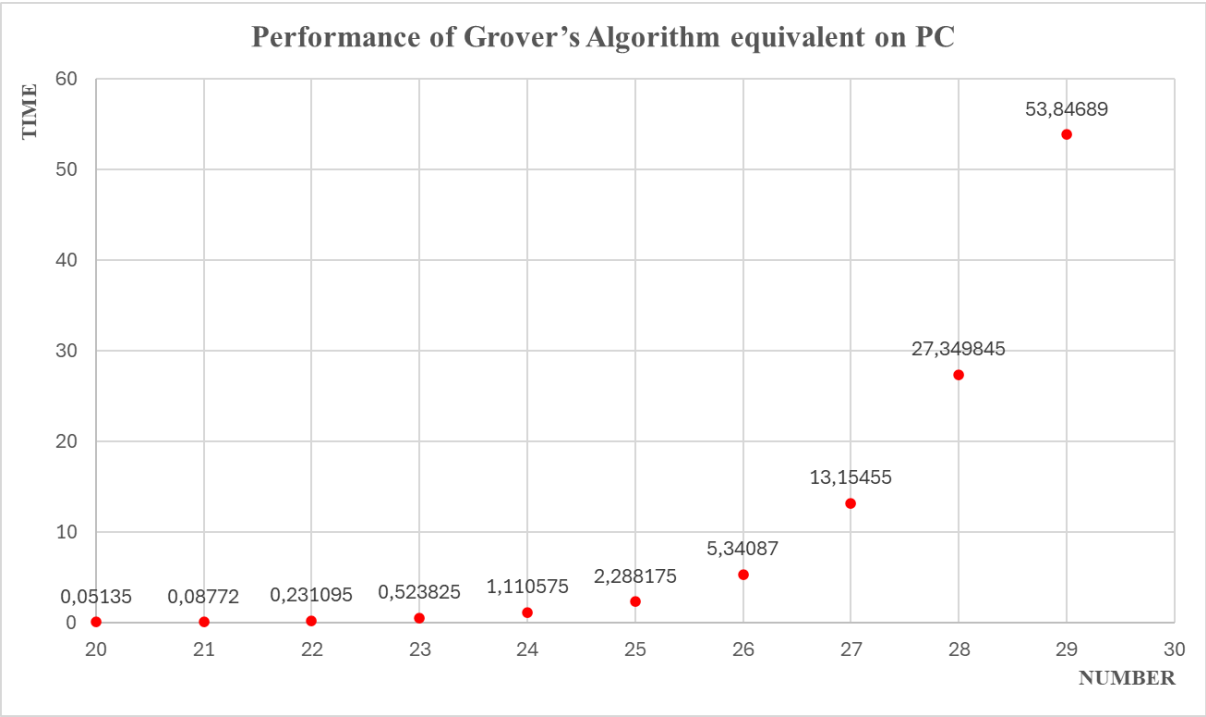


Figure 10.2: Performance of Grover's Algorithm equivalent on PC (Created by the Author)

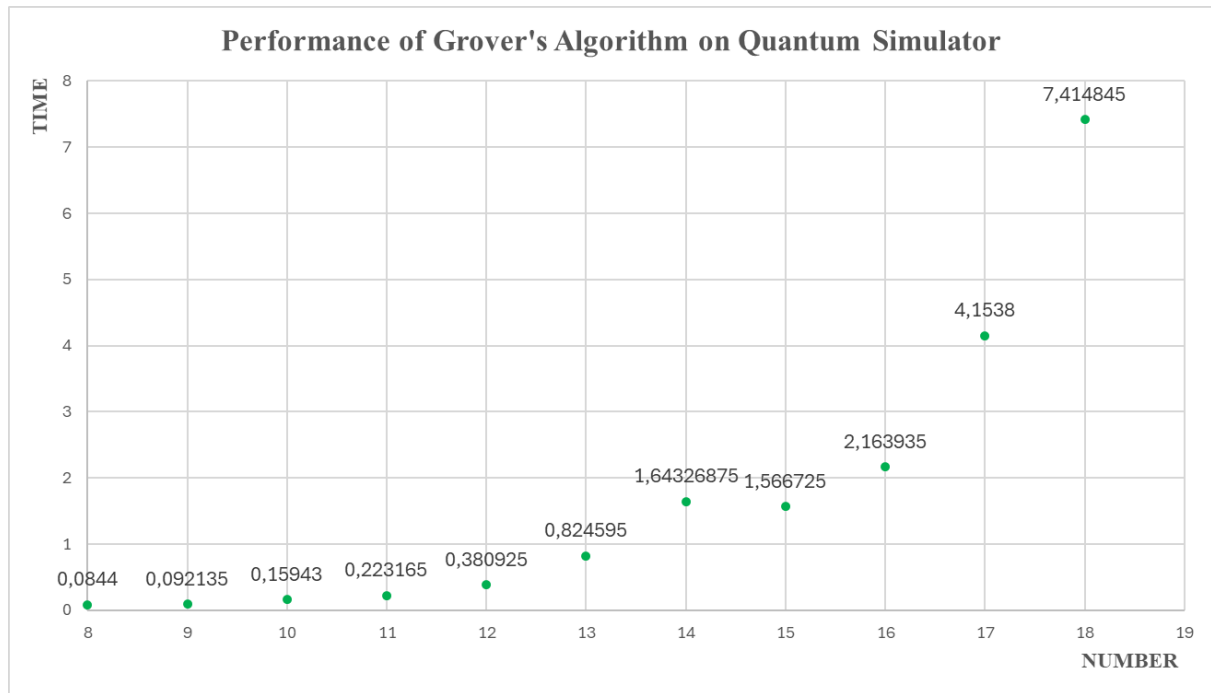


Figure 10.3: Performance of Grover's Algorithm on Quantum Simulator (Created by the Author)

11.3. Results Discussion

The research was conducted on all three systems, meaning a classical computer, a quantum simulator, and a quantum computer. The results were presented only for classical calculations (Figure 10.2) and simulated (Figure 10.3), for the same reason that they were omitted when testing the implementation of Shor's algorithm.

The graphs clearly show that reality matches expectations (Figure 10.1), the data forms an exponential hyperbola. It can also be seen that for some values, the graph deviates from the norm - this is caused by processor clocking disruptions or momentary drops in computer power due to current loads.

11.4. Real-world Applications

Grover's algorithm has significant potential in practical applications, which is used in many fields of science. The use of the algorithm in searching large data sets is important in areas such as database search, optimization, cryptography, machine learning, and pattern recognition.

11.4.1. Database Search and Optimization

Grover's algorithm is used in database searches, enabling faster retrieval of information from unsorted data sets. This has great implications for data analysis, information retrieval systems, and data-driven decision-making. The algorithm can also support solving optimization problems, such as finding the best solution among many possibilities, offering a quantum advantage in efficiently finding optimal solutions.

11.4.2. Cryptography

Grover's algorithm has potential impact on breaking symmetric encryption schemes. In theory, it accelerates brute-force attacks by searching for encryption keys with quadratic speedup compared to classical methods. This highlights the need to develop quantum-resistant cryptographic algorithms to reduce the risk associated with such attacks.

11.4.3. Machine Learning and Pattern Recognition

Another application of the described algorithm is its use in solving machine learning problems such as classification, clustering, and pattern recognition. It offers a potential advantage in tasks involving finding and matching patterns in large data sets.

11.4.4. Development of quantum algorithms [\[12\]](#)

The development of more complex quantum algorithms can be based on Grover's algorithm. Its concepts and methods are used in many other quantum algorithms, such as Quantum Amplitude Amplification and Quantum Walk Search, which contributes to the development of the entire field of quantum computing.

Although Grover's algorithm shows impressive capabilities and finds applications in many scientific fields, it is important to understand its limitations. Unlike some quantum algorithms, such as Shor's algorithm described above, Grover's algorithm does not offer exponential speedup. It should also be noted that the program cannot be parallelized. To ensure the best results, calculations must be performed sequentially, which translates into the competitiveness of multi-threaded algorithms that allow for the use of this computational advantage. Therefore, it is crucial to choose the right algorithm for a specific problem to get the best results.

12. Public-key cryptography - Explaining key rules of public-key cryptography, in context of Quantum Key Distribution.

Public keys dominate today's world, becoming widely used, among other things, in Internet security. Until now, nothing has undermined their security, but with the development of quantum computers, the increase in their capabilities - commonly used public keys are becoming vulnerable to attacks using algorithms based on quantum computing. The hope for the development of secure communications is brought about by a technology called Quantum Key Distribution [\[58\]\[59\]\[60\]](#).

12.1. Public key

12.1.1. Introduction to public key cryptography

Public key cryptography, also known as asymmetric cryptography, is one of the most important developments in modern cryptology. Its fundamental feature is the separation of the cryptographic key into two parts: the public key, which can be freely shared, and the private key, which remains secret and known only to the owner. This asymmetry makes it possible to perform operations such as encryption, digital signatures, and key exchange in a way that ensures the security of communications in the presence of potential adversaries. However, in the context of the rapid development of quantum computers, there are new challenges related to the security of public key cryptography. One possible solution to these problems is Quantum Key Distribution (QKD).

12.1.2. The principles of public key cryptography

Public key cryptography is based on the use of difficult mathematical problems that are easy to solve in one direction (encryption with a public key) but difficult in the other (decryption without knowledge of the private key). Among the most important algorithms used in asymmetric cryptography are:

- **RSA algorithm:** It is based on the problem of factorization of large numbers, where the public key is the product of two large prime numbers and the private key is the respective inverses of their Euler functions. It has already been described in detail in the chapter discussing the Shor algorithm.
- **ElGamal algorithm:** It is based on the difficulty of solving the discrete logarithm problem, which is a straightforward operation in one direction (cyclic group triplication), but extremely difficult in the opposite direction.
- **Elliptic curves (ECC):** These take advantage of the algebraic properties of elliptic curves over finite bodies, offering a similar level of security to RSA with much smaller key sizes.

These algorithms are the basis of modern cryptographic protocols, such as SSL/TLS, PGP, and digital signatures. In practice, public key cryptography enables the secure exchange of information, even if the communication channel is not secure.

12.1.3. Applications of public key cryptography

Public key cryptography is used in many areas, including:

- **Secure communications:** Encrypting messages using a public key allows them to be transmitted securely, as only the holder of the appropriate private key can decrypt them.
- **Digital signatures:** A private key is used to create a digital signature that can be verified with a public key, ensuring the integrity and authenticity of messages.
- **Public key infrastructure (PKI):** Supports the management of public keys, enabling them to be shared securely and verify the identity of entities using asymmetric cryptography.
- **SSL/TLS:** Provides secure Internet connections, using public key cryptography to exchange symmetric session keys, which are then used to encrypt data transmissions.
- **Blockchain systems and cryptocurrencies:** Use asymmetric cryptography to secure transactions and manage user identities.

12.2. Quantum Key Distribution (QKD)

12.2.1. Public key cryptography in the face of quantum computers

The development of quantum computers poses a serious threat to traditional public key cryptography systems. Shor's algorithm, proposed in 1994, makes it possible to factorize large numbers and solve the discrete logarithm problem in polynomial time, potentially breaking RSA, ElGamal and elliptic curve-based cryptography. Faced with these threats, the research community has focused on finding new methods to ensure the security of communications in the quantum age.

One such solution is **Quantum Key Distribution (QKD)**, which uses the principles of quantum mechanics to provide unconditionally secure cryptographic key exchange. QKD, unlike classical asymmetric cryptography, is not based on computational difficulties, but on fundamental principles of quantum physics, such as the Heisenberg principle and the principle of impossibility of quantum cloning.

12.2.2. Principles of Quantum Key Distribution

QKD allows two parties (for the purpose of the example, call them Alice and Bob) to securely exchange a cryptographic key over a quantum channel that may be susceptible to eavesdropping. The basic QKD protocol is the **BB84 protocol**, developed by Charles Bennett and Gilles Brassard in 1984. This protocol uses single photons that can be prepared in different quantum states, representing bits of information (zero or one). The key elements of QKD operation are:

- **Superposition principle:** A photon can be prepared in one of four states (horizontal, vertical, left-handed, or right-handed), and its exact state is unknown until the measurement.
- **The principle of measurement unpredictability:** Alice and Bob's choice of measurement base is random. Measuring at the wrong base introduces an error to detect the presence of eavesdropping.
- **Eavesdropping detection:** If a third party (Eve) tries to eavesdrop on the quantum channel, it introduces interference, which is detectable by Alice and Bob by comparing parts of the exchanged data.

Once the key exchange is complete, Alice and Bob can use classical error correction and privacy algorithms to obtain a shared secret key, which can then be used to encrypt messages using classical cryptographic methods such as AES.

The BB84 protocol ensures that any eavesdropping attempt changes the quantum state of the particles, thus informing the communicating parties of the presence of an intruder. This inherent security feature offers a robust solution for secure key exchange, making QKD a key area of research in the quantum cryptography field.

12.2.3. Applications and the future of QKD

QKD is already being used in some critical applications, such as securing government, financial and military communications. The main advantages of QKD are:

- **Unconditional security:** QKD security does not depend on computational difficulties, but on fundamental laws of physics.
- **Immunity to quantum attacks:** QKD is one of the few solutions that remains secure even in the face of a quantum computer.

However, QKD also has its limitations, which may hinder its widespread use:

- **Range and throughput:** Current QKD systems have limited range (tens to hundreds of kilometers) and low throughput, which may limit their utility on a wide scale.
- **Hardware requirements:** QKD requires specialized hardware, such as single photon sources and high-sensitivity detectors, which increases the cost and complexity of implementation.

- **Integration with existing systems:** Implementing QKD requires compatibility with existing cryptographic infrastructure, which can be challenging.

Despite these limitations, developments in quantum technologies, including quantum networks and satellite-based QKD systems, could expand the reach and availability of this technology in the future, making it a key component of the global security infrastructure.

12.2.4. Technologies used in the implementation of QKD.

It is worthwhile to introduce the technologies and devices used to implement QKD. The discussion should include:

- **Photon sources:** Essential for generating single photons, which are the carriers of information in QKD. Today's photon sources are based on semiconductor and cascade lasers capable of emitting single photons on demand.
- **Photon detectors:** Used to detect the quantum states of photons. Modern detectors, such as semiconductor-based avalanche detectors, can record single photons with high efficiency and low noise.
- **Error correction and privacy systems:** After exchanging keys through the quantum channel, Alice and Bob use classical algorithms for error correction and procedures for privacy enhancement, resulting in a common cryptographic key.

12.2.5. Challenges and limitations in QKD implementation

Despite QKD's enormous potential, there are practical challenges that may limit its widespread use:

- **Transmission range:** In current QKD systems, transmission range is limited to tens or hundreds of kilometers due to signal attenuation in optical fibers and losses in the atmosphere. Research on quantum amplification and quantum repeaters may increase this range in the future.
- **Integration with existing networks:** Integrating QKD with existing communications infrastructures, such as the Internet, requires the development of compatible protocols and technologies that allow the secure transmission of quantum keys.
- **Implementation costs:** QKD requires sophisticated hardware and expertise, which can raise the cost of implementing and maintaining such systems.

12.2.6. QKD protocols: Extension beyond BB84

Although BB84 is the most popular QKD protocol, there are many other protocols that offer various levels of security and efficiency:

- **E91 protocol:** based on quantum entanglement, where two entangled photons are sent to Alice and Bob. This protocol offers enhanced security compared to BB84.
- **BBM92 protocol:** Also based on quantum entanglement, but simpler to implement compared to E91, with similar security properties.

12.2.7. The future of cryptography in the era of quantum computing

Compare and analyze potential scenarios for the future of cryptography in the face of quantum computers, where QKD is one solution, but not the only one:

- **Post-quantum cryptography:** cryptographic algorithms resistant to quantum attacks, such as lattice-based cryptography, could be an alternative or complement to QKD.
- **Quantum networks:** The development of quantum networks that will enable broadband quantum communications and key distribution on a global scale, using satellite-based QKD systems.

Summary

Public key cryptography is a pillar of modern cybersecurity, enabling the secure exchange of information in insecure environments. However, the development of quantum computers poses a serious threat to its existing performance, forcing the search for innovative solutions such as Quantum Key Distribution. QKD, using the principles of quantum mechanics, offers unconditional security, but requires further research and development to become a widely used technology. As quantum computers and quantum networks advance, QKD could play a key role in the future of cryptography and securing digital communications.

13. Post-Quantum Cryptography - Challenges for contemporary and future cryptographic.

Introduction

Cryptography is the cornerstone of security in a modern digital society. Cryptographic systems secure communications, protect data from unauthorized access and enable secure financial transactions. Traditional cryptographic systems, such as RSA, DSA or ECC, base their security on the complexity of mathematical problems such as factorization of large numbers or discrete logarithms. The introduction of quantum computers, capable of solving these problems exponentially faster than classical computers, poses a serious threat to these traditional systems. In response to this threat, post-quantum cryptography is evolving to develop algorithms that are resistant to quantum attacks [\[21\]\[22\]\[23\]](#).

13.1. Post-quantum cryptography

13.1.1. Operating principles & examples

Post-Quantum Cryptography (PQC) includes cryptographic algorithms that are resistant to both classical and quantum attacks. Unlike quantum cryptography, which uses the principles of quantum mechanics, PQC is based on mathematical problems that remain difficult to solve even with quantum computers. The major classes of mathematical problems on which PQC is based include:

1. **Lattice-based Cryptography (LQC)** problem: Lattice-based cryptography is one of the most promising areas of PQC. Grids are multidimensional geometric structures, and the security of grid-based algorithms stems from the difficulty of finding the shortest vector in each grid (**SV**P, for Shortest Vector Problem) or solving the Closest Vector Problem (**CV**P). Grid-based algorithms, such as NTRUEncrypt, Kyber, and Dilithium, show high resistance to quantum attacks.
2. **Code-based Cryptography:** These algorithms are based on the problem of decoding random linear codes, which is difficult for both classical and quantum computers. Code based on McEliece's problem is one of the oldest and best-known examples of this class of algorithms.
3. **Multivariate Quadratic Equations (MQ) Cryptography:** Algorithms in this group are based on the problem of solving systems of quadratic equations over finite bodies. This problem is known for its computational difficulty, which makes it resistant to quantum attacks. Rainbow family algorithms are examples of such algorithms.

4. **Hash-based Cryptography:** In this field, cryptographic security is based on the difficulty of inverting one-way hash functions. An example is the SPHINCS algorithm, which provides resistance to quantum attacks using Merkle trees.
5. **Isogeny-based Cryptography:** Isogeny-based algorithms between elliptic curves exploit the difficulty of finding isogeny between two curves. An example is SIKE (Supersingular Isogeny Key Encapsulation), which is currently one of the most advanced approaches in post-quantum cryptography.

Summarizing the results of the competition described in the chapter on the history of Post Quantum algorithms, after several years of intensive research and evaluation, NIST announced four algorithms as finalists to be standardized in 2022 [\[23\]](#):

- **Crystals-Kyber (Network-based Cryptography):** An algorithm for key exchange. It is one of the most promising algorithms, considered very secure and efficient. Its security is based on the Learning With Errors (LWE) problem, and applications include asymmetric encryption and key exchange.
- **Crystals-Dilithium (Network-Based Cryptography):** A cryptographic algorithm for generating digital signatures based on the Short Integer Solution (SIS) problem in networks. also showing great promise in terms of security and performance.
- **Falcon (Network-based Cryptography):** Another algorithm for digital signatures that features high performance with adequate security.
- **SPHINCS+ (Cryptography based on hash functions):** An algorithm for digital signatures, notable for its special resistance to diverse types of attacks, albeit at the expense of larger signature sizes.

13.1.2. Applications

Post-quantum cryptography will find application in many areas where long-term data protection is required:

1. **Communication security:** Information exchanges over the Internet, including TLS/SSL protocols, must be resistant to quantum attacks to ensure the security of communications. Introducing PQC into these protocols will be key to protecting future data transmissions.
2. **Financial systems:** Banks and financial institutions that store vast amounts of sensitive data need to implement PQC to protect against future quantum attacks that could break classic encryption algorithms.

3. **Critical infrastructure:** Power systems, communications and other critical infrastructures need security that is resistant to quantum attacks to prevent the possibility of unauthorized actors taking control.
4. **Long-term data storage:** Information that must be protected for decades, for instance medical records, or government data, must be secured against future threats from quantum computers.

13.1.3. Challenges and difficulties in implementation

Although post-quantum cryptography is a promising field, it also faces numerous challenges:

1. **Algorithm optimization:** Post-quantum algorithms often require more computational resources compared to classical algorithms, which can lead to performance issues, especially in resource-constrained environments such as IoT devices.
2. **Standardization:** the National Institute of Standards and Technology (NIST) PQC standardization process is key to the widespread adoption of new cryptographic standards. However, choosing algorithms that are secure yet efficient is a challenge, given the variety of potential applications.
3. **Compatibility with existing infrastructure:** Converting current systems to be compatible with PQC can be costly and complicated. This requires both software and hardware upgrades.
4. **Protection against side-channel attacks:** PQC algorithms can be vulnerable to side-channel attacks that exploit information leaked during computation, such as energy consumption or electromagnetic radiation.

The field of quantum-resistant cryptography focuses on developing cryptographic algorithms that are resistant to quantum attacks while being compatible with existing communications infrastructure. Algorithms, especially those mentioned above, aim to provide security in a post-quantum world. The transition to PQC is essential for maintaining the confidentiality and integrity of sensitive information in an era when quantum computers will become operational.

13.1.4. The future

The future of post-quantum cryptography involves dynamic developments in both theory and practice. This is expected over the next few decades:

1. **Increase the share of PQC:** As standards stabilize and algorithms become more optimized, widespread adoption of PQC is expected in many sectors, from banking to mobile technologies.
2. **Integration with quantum cryptography:** PQC can be used in conjunction with quantum cryptography (e.g., Quantum Key Distribution) to provide an even higher level of security. Quantum cryptography can provide real-time security, while PQC can protect data long-term.

3. **Development of new algorithms:** Research into new mathematical problems that may be difficult to solve even with quantum computers will continue, which could lead to the development of new, more efficient algorithms.
4. **Hardware technology improvements:** As computer technology develops, including hardware specifically designed to manage PQC, the performance and efficiency of these algorithms will improve.

Summary

Post-quantum cryptography is an indispensable tool for protecting data in the coming era of quantum computers. While it faces many challenges, such as algorithm optimization, standardization, and integration with existing systems, its importance will grow as technology advances. The introduction of PQC into widespread use is key to ensuring data security in the coming decades, both in terms of defending against potential threats from quantum computers and in the face of ever-evolving cyber threats. Therefore, further research, technology development and international cooperation are essential to meet the challenges posed by the future of cryptography.

14. Conclusions

14.1. Conclusions and research results. Document structure and its individual elements. Current state of the field.

14.1.1. Limitations of Modern Quantum Computers - Assessment of the Current State of Quantum Technology

Modern quantum computers, while representing a technological breakthrough, still face significant obstacles that limit their practical application. One key problem is the high susceptibility to quantum errors, resulting from the fragile nature of quantum states and their sensitivity to environmental perturbations. Although systems such as IBM's one hundred twenty-seven qubits Eagle processor represent a step forward, the current state of technology still does not allow for the widespread implementation of advanced quantum algorithms in real-world scenarios. An example of this is Shor's algorithm, which, although theoretically capable of factoring large integers, can only manage remarkably simple examples such as the number twenty-one on today's quantum computers. Limited scalability and difficulties associated with quantum error correction pose serious challenges that must be overcome before quantum computers can realize their full potential, especially in the context of breaking modern cryptographic algorithms. Current research efforts are focused on developing more advanced quantum technologies that can address these challenges.

14.1.2. Results of comparing quantum algorithms with classical equivalent

The conducted research compared the performance of quantum algorithms – Shor's and Grover's – with their classical equivalents, which allowed for identifying key differences between these approaches. The results clearly indicate the advantage of quantum algorithms in terms of computational complexity, especially in the case of Shor's algorithm, which shows logarithmic scalability compared to the exponentially increasing complexity of classical algorithms. Nevertheless, in current technological conditions, both algorithms, both Shor's and Grover's, although theoretically enabling effective factoring of numbers or searching extensive, unsorted databases, in practice, using available technologies, can only be applied to remarkably simple research cases, even demonstration/learning purposes. Tests conducted on both the IBM quantum computer and the simulator of such a computer confirmed that quantum algorithms are indeed more effective, but their full potential remains unattainable due to current technological limitations. As a result, although quantum algorithms offer the prospect of significantly accelerating computations in the future, classical algorithms currently remain more practical and efficient for many applications.

14.1.3. The interdisciplinary nature of quantum technology research

Quantum technology research has a clearly interdisciplinary character, combining elements from fields such as physics, mathematics, computer science, and cryptography. Contemporary work on quantum algorithms, such as the one conducted in this work, requires a deep understanding of quantum mechanics, which is the foundation for interpreting and understanding algorithms such as Shor's or Grover's algorithms. At the same time, implementing these algorithms on quantum computers requires knowledge of computer engineering and programming in specific quantum languages, such as Qiskit. In turn, mathematics, in particular number theory and linear algebra, plays a key role in the formal description and analysis of these algorithms, and cryptography in the study of their potential practical applications. Such cooperation between different fields of science not only expands the boundaries of knowledge, but also enables the creation of new, innovative solutions. This work is therefore an example of constructive collaboration that can lead to considerable progress in the development of quantum technologies, showing how interdisciplinarity is an indispensable element of effective research in this dynamically developing field. Quantum mechanics, cryptography, computer science, and quantum engineering are linked, and progress in each of these fields contributes to the development of the entire technology. This work can be a valuable source of information for both physicists, dealing with theoretical foundations, and for mathematicians and computer scientists working on cryptography and implementation of algorithms on quantum computers.

14.1.4. Difficulties in implementing quantum algorithms - Technological and Implementation Challenges, Quantum Software Challenges, Role of Expert Support and Cooperation with Research Institutions

The implementation of quantum algorithms, such as the Shor and Grover algorithm, encounters **numerous technological and programming challenges** resulting from the current state of development of quantum technology. Despite the use of advanced hardware, such as the one hundred twenty-seven qubits Eagle processor from IBM, difficulties related to precise control of quantum states, decoherence and quantum errors, constitute significant limitations in achieving the expected results. In addition, the implementation of these algorithms required deep knowledge of both quantum computing and the ability to adapt to dynamically changing programming tools. **The Qiskit software** deserves special attention, which is being dynamically developed and was used to conduct tests in this work. Unfortunately, with its changes, additional problems arise in the form of compatibility of existing solutions with the latest versions of the software. Therefore, the support of experts from the **Poznan Supercomputing and Networking Center (PSNC)** turned out to be crucial for the success of the project, emphasizing the importance of cooperation with research institutions and the need for continuous updating of knowledge in this area. Contemporary research on quantum algorithms

therefore still requires an interdisciplinary approach, integrating knowledge from different fields and close cooperation between scientists and engineers to effectively overcome current technological limitations and implement theoretical concepts in practice.

14.1.5. Development of hardware technologies and the foundations of quantum mechanics

In this paper, special attention is paid to discussing the fundamentals of quantum mechanics, which are the foundation for understanding the operation of quantum computers and their algorithms. Quantum mechanics, with its complex principles, phenomena, and concepts, provided an essential theoretical background for the implementation and analysis of Shor and Grover's algorithms. A thorough explanation of these topics was crucial to provide the reader with a solid understanding of the basis of all quantum technology.

The paper also discusses several types of quantum computers, with a special emphasis on those based on superconducting qubits and gate technology, such as the IBM Eagle processor, which was used to implement selected algorithms. Understanding how these computers work, including the specifics of quantum operations and the limitations resulting from the current state of technology, was crucial to assessing the practicality of quantum algorithms compared to their classical counterparts.

The development of hardware technologies, which was also thoroughly analyzed, has a significant impact on the possibilities of practical implementation of quantum algorithms. As technology advances, the potential of quantum computers increases, which makes further research in this field extremely promising. In the context of this work, these developments highlighted both the current challenges and prospects in the field of quantum computing.

14.1.6. Predictions vs. Actual Research Results

The predictions about the efficiency of quantum algorithms, especially Shor's and Grover's, were reflected, especially in the context of quantum simulations. The results obtained on quantum computer simulators were dependable and consistent with the theoretical assumptions. Shor's algorithm, which in theory is characterized by logarithmic complexity, indeed showed such efficiency in simulations, which confirms that the basic theoretical assumptions were correctly realized.

Similarly, Grover's algorithm, whose complexity should be the square root of the exponential, also confirmed its theoretical properties during tests on simulators. These results indicate that despite the limitations of current quantum technology, the theoretical foundations of both algorithms are solid, and their potential is real.

However, when these algorithms were implemented on real quantum computers, significant differences appeared. In practice, Shor's algorithm was only able to factor small numbers, and full use of its potential requires further development of hardware technologies. Grover's algorithm, although effective in theory, has encountered similar technological limitations that prevent its full application in practice. The research results clearly show that the development of quantum technologies is crucial for the future success of these algorithms.

During the research work, many expectations were set for the capabilities of quantum algorithms, especially in the context of their theoretical advantage over classical algorithms. However, predictions regarding their efficiency turned out to be significantly ahead of the current state of quantum technology. Despite considerable progress, actual results have indicated limitations that currently prevent the practical application of this algorithm to breaking cryptography on an industrial scale.

14.2. Implications for further research

14.2.1. Quantum Hardware Development Prospects [\[20\]\[61\]\[62\]](#)

Plans for the development of quantum computers are no secret. IBM, as one of the leaders in this field, in its quantum computer development strategy (IBM Quantum roadmap) announced the introduction of new generations of quantum processors by 2025, which are to significantly increase computing capabilities. These plans include the development of processors with thousands of qubits and the implementation of advanced error correction techniques, which are key to increasing the reliability of quantum computing. By delving into the details, it is possible to reach specific information. In its plan for the development of quantum technology by 2025, IBM announces considerable progress in the construction of quantum computers. Plans for the near future include the introduction of the four hundred sixty-two qubits “Flamingo” processor with a built-in quantum communication link, and then the release of a demonstration of this architecture by linking together at least three Flamingo processors into a one thousand three hundred eighty-six qubits system. The result of these advances will be the introduction of the “Kookaburra” processor. Kookaburra will be a one thousand three hundred eighty-six qubits multi-chip processor with a quantum communication link. As a demonstration, three Kookaburra chips will connect into a four thousand one hundred fifty-eight qubits system connected by quantum communication for our users.

IBM plans to scale quantum systems by connecting quantum processors to classical CPUs and GPUs and developing a software infrastructure that will enable the integration of quantum and classical computations in a seamless and efficient way.

The consequence of the systematic development of this technology, after 2033, will be a quantum supercomputer equipped with thousands of qubits capable of handling one billion gates, which will allow the full power of quantum computation to be used. The goal will be a one hundred thousand qubits Blue Jay system that will define two thousand qubits capable of managing a total of one billion gates. The middleware will integrate this system with increasingly powerful quantum-centric supercomputers.

14.2.2. Quantum Algorithm Optimization, Potential of Quantum, and Post-Quantum Algorithms in Cryptography. Future Applications

Quantum algorithm optimization and the potential of quantum and post-quantum algorithms in cryptography are key issues in the development of quantum technology. This paper analyzes the capabilities of algorithms such as Shor's and Grover's, which have the potential to revolutionize modern cryptographic systems. Shor's algorithm, although currently limited by available hardware resources, has the potential to break standard cryptographic algorithms such as RSA. In turn, Grover's algorithm allows for accelerating the search of large data spaces, which can be used to strengthen security mechanisms.

In the context of advances in quantum technology, especially in hardware, the optimization of these algorithms becomes not only possible but also necessary. The prospects for the development of quantum hardware, such as IBM's plans, described in the previous chapter, open new possibilities for the implementation of advanced algorithms. Future research should focus on further optimization of these algorithms and their integration with developing quantum technologies, which may bring a breakthrough in the field of cryptography and data security in the coming years.

14.2.3. The importance of the work for future research, The importance of the work for education and popularization of knowledge

This work has significant implications for both future research, education, and popularization of knowledge in the field of quantum algorithms and quantum technology. The conclusions and results contained in it can form the basis for further development of efficient methods of quantum computation and post-cryptography, thus supporting the development of these fields. Thanks to the in-depth explanation of the foundations of quantum mechanics and quantum algorithms, this work can serve as a valuable source of knowledge for students, researchers, and people interested in quantum technologies. Popularization of knowledge about these advanced technologies can contribute to greater understanding and interest among the wider scientific community, as well as support education at various levels of advancement, from academic to general society.

Thus, this work not only develops knowledge about the current state of research, but also inspires further search for innovative solutions and innovations in this dynamically developing field.

Final conclusion

It is time to answer the questions asked in the Introduction chapter.: Are the secrets stored on the Internet still safe? Are the logins, passwords, accounts, or even digital signatures used and will they remain protected? Can they be cracked, stolen, or even bypassed in a matter of minutes, contrary to existing knowledge and methods? Will the cryptographic systems, encryption protocols, authentication mechanisms used so far become obsolete and insufficiently secure in the face of rapidly developing quantum algorithmics? Is there already a demand for alternative methods?

Currently, secrets stored on the Internet, such as logins, passwords, accounts or digital signatures, are protected by classical cryptographic methods, which are based on mathematical difficulties such as prime number factorization or the discrete logarithm problem, so it can be assumed that they are still safe, and the logins or passwords used are protected. Although the current technology does not threaten them, with the development of quantum computers and algorithms, and particularly Shor's algorithm, this situation may change. In the future, when quantum computers reach the appropriate power, they can break classical algorithms in a brief time, which will make the current cryptographic systems outdated and insufficiently secure. Therefore, it is necessary to develop alternative post-quantum cryptography methods now, which will be resistant to attacks using quantum computers. Work on such solutions is in progress, which allows humanity to prepare for future threats and secure information against potential quantum attacks, to ensure security before what is yet to come.

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